

SS1 PHYSICS LESSON NOTES

FIRST TERM

1. Fundamentals and Derived Quantities and Units
2. Measurements of mass, weight, length and time
3. Position, Distance and Displacement
4. Scalars and Vectors
5. Speed, Velocity and Acceleration
6. Motion



WEEK 1 - FUNDAMENTAL AND DERIVED QUANTITIES AND UNITS

DEFINITION OF PHYSICAL QUANTITIES

Physical quantities are properties of materials or systems that can be measured or quantified. They are classified into fundamental and derived quantities.

FUNDAMENTAL QUANTITIES AND UNITS

Fundamental quantities are independent physical quantities that cannot be expressed in terms of other quantities. They serve as the foundation for deriving other physical quantities.

List of the Seven (7) Fundamental Quantities and Their SI Units

Fundamental Quantity	SI Unit	Symbol	Measuring Instrument
Length	metre	m	Ruler, Meter rule
Mass	kilogram	kg	Beam balance, Scale
Time	second	s	Stopwatch, Clock
Electric Current	ampere	A	Ammeter
Temperature	kelvin	K	Thermometer
Amount of Substance	mole	mol	Chemical balance
Luminous Intensity	candela	cd	Photometer

Characteristics of Fundamental Quantities:

- They are independent (cannot be derived from other quantities).
- They have standard SI units.
- They are used to define derived quantities.

DERIVED QUANTITIES AND UNITS

Definition: Derived quantities are dependent physical quantities obtained by combining fundamental quantities through mathematical operations (multiplication, division, exponents, etc.).

Examples of Derived Quantities, Their Formulas, and SI Units

Derived Quantity	Formula (Derivation)	SI Unit	Symbol (in Fundamental Units)
Area	Length $\times$ Width	m ²	m ²
Volume	Length $\times$ Width $\times$ Height	m ³	m ³
Density	Mass / Volume	kg/m ³	kgm ⁻³
Velocity	Displacement / Time	m/s	ms ⁻¹
Acceleration	Change in Velocity / Time	m/s ²	ms ⁻²
Force	Mass $\times$ Acceleration	newton (N)	kgms ⁻²
Work/Energy	Force $\times$ Distance	joule (J)	kgm ² s ⁻²
Power	Work / Time	watt (W)	kgm ² s ⁻³
Pressure	Force / Area	pascal (Pa)	kgm ⁻¹ s ⁻²

Characteristics of Derived Quantities:

- They depend on fundamental quantities.
- Their units are combinations of fundamental units.
- They can be expressed in terms of dimensions.

DIMENSIONS OF PHYSICAL QUANTITIES

Definition: Dimensions represent the powers to which fundamental quantities (Mass [M], Length [L], Time [T], etc.) are raised in a derived quantity.

Dimensional Formulas for Some Quantities

S/N	Quantity	Formula	Dimensional Formula
1	Velocity (v)	Displacement $\div$ Time	[L][T ⁻¹]
2	Acceleration (a)	Velocity $\div$ Time	[L][T ⁻²]
3	Force (F)	Mass $\times$ Acceleration	[M][L][T ⁻²]
4	Work (W)	Force $\times$ Distance (Displacement)	[M][L ²][T ⁻²]

Uses of Dimensional Analysis:

- Checking the **correctness** of equations.
- Deriving relationships between physical quantities.
- Converting units from one system to another.

IMPORTANCE OF UNITS AND MEASUREMENTS IN PHYSICS

1. **Standardization:** Ensures uniformity in measurements worldwide.
2. **Accuracy:** Reduces errors in scientific experiments.
3. **Reproducibility:** Allows experiments to be repeated with the same results.
4. **Communication:** Enables scientists to share data universally.

Unit Conversion

Students should know how to convert between units.

Examples:

1. Convert 100 cm to metres:
 $100\text{ cm} = 100\text{ cm} = 100 \div 100 = 1\text{ m}$
2. Convert 2 hours to seconds:
 $2\text{ hr} = 2 \times 60 \times 60 = 7200\text{ s}$

Prefix	Symbol	Multiplier
Kilo	K	10^3
Centi	C	10^{-2}
Milli	M	10^{-3}
Micro	μ	10^{-6}
Mega	M	10^6

EVALUATION QUESTIONS

1. Define fundamental quantities and give **three** examples.
2. Differentiate between fundamental and derived quantities.

3. Derive the SI unit of **force**.
4. What is the dimensional formula for **work**?
5. Why is standardization important in measurements?

ASSIGNMENT

1. List **five** derived quantities, their formulas, and SI units.
2. Derive the dimensional formula for **pressure**.
3. Explain why **time** is a fundamental quantity but **velocity** is derived.



WEEK 2 - MEASUREMENT OF MASS, WEIGHT, LENGTH, AND TIME

MEASUREMENT OF MASS

Definition: Mass is the amount of matter contained in a body. It is a **scalar quantity** (has magnitude only). SI Unit is in **kilogram (kg)**

Instruments Used:

- Beam balance
- Triple beam balance
- Electronic/digital balance
- Chemical balance (for small masses)

Example:

Mass of a bag of rice = 25 kg

MEASUREMENT OF WEIGHT

Definition: Weight is the force with which the earth attracts a body. It is a **vector quantity** (has both magnitude and direction). SI Unit is in **newton (N)**.

Formula:

Weight (W) = Mass (m) × Acceleration due to gravity (g)

Where: $g = 9.8 \text{ m/s}^2$ on Earth

Instruments Used:

- Spring balance
- Newton meter

Example:

Mass = 5 kg

Weight = $5 \times 9.8 = 49 \text{ N}$

Note:

Weight varies from place to place due to changes in gravitational field, while mass remains constant.

MEASUREMENT OF LENGTH

Definition: **Length** is the distance between two points. It is a **scalar quantity**. SI Unit is in **metre (m)**.

Instruments Used:

Instrument	Measurement Range	Least Count (Accuracy)
Ruler / meter rule	Up to 1 metre	1 mm (0.1 cm)
Tape measure	Up to 50 metres	1 mm
Vernier caliper	Up to 15 cm	0.01 cm (0.1 mm)
Micrometer screw gauge	Up to 2.5 cm	0.01 mm

Precautions:

- Take reading at eye level to avoid **parallax error**.
- Ensure instrument is clean and properly calibrated.

Unit Conversions:

- 1 km = 1000 m
- 1 m = 100 cm = 1000 mm
- 1 inch = 2.54 cm

MEASUREMENT OF TIME

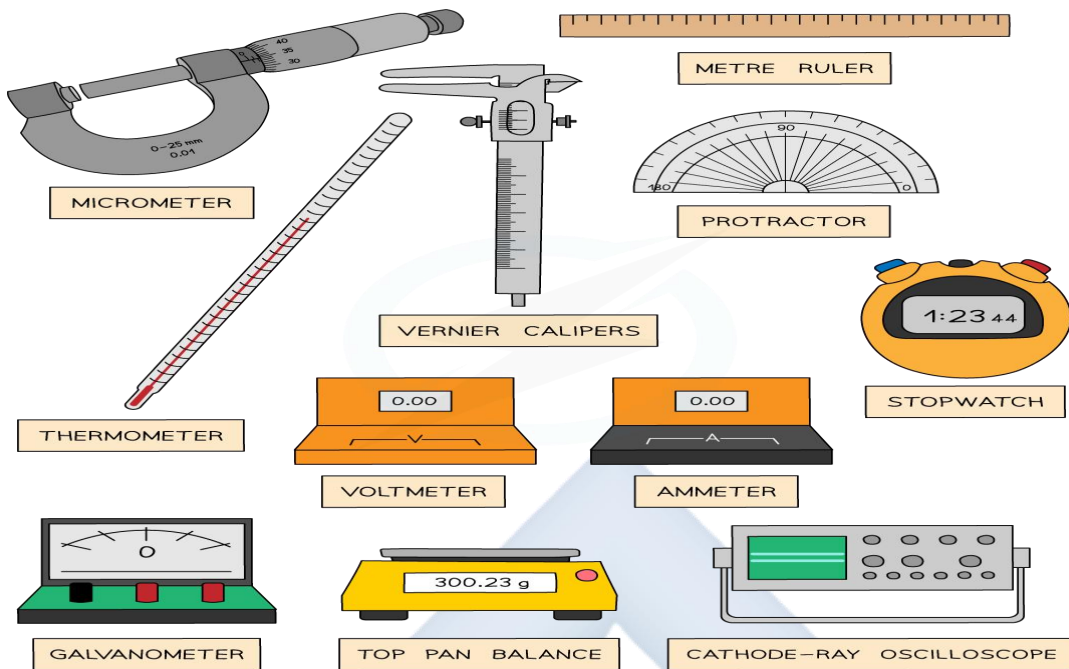
Definition: **Time** is the ongoing sequence of events taking place. It is a **scalar quantity**. SI Unit is in **second (s)**.

Instruments Used:

- Clock
- Stopwatch (manual or digital)
- Pendulum clock (for experiments)

Unit Conversions:

- 1 minute = 60 seconds
- 1 hour = 3600 seconds
- 1 day = 86,400 seconds



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COMMON ERRORS IN MEASUREMENT

- **Parallax Error:** Mistake from looking at scale from the wrong angle.
- **Zero Error:** Instrument does not start from zero.
- **Instrumental Error:** Faulty or miscalibrated equipment.
- **Human Reaction Time:** Delay in starting/stopping stopwatch.

Importance of Measurement in Physics

- Provides standardization and accuracy in experiments.
- Essential for comparing and predicting physical phenomena.
- Used in industries, engineering, construction, medicine, etc.

EVALUATION:

1. Define the following term:

- (a) Mass
 - (b) Weight
 - (c) Length
 - (d) Time
2. State the SI units for:
- (a) Mass,
 - (b) Weight,
 - (c) Length,
 - (d) and Time
3. List two instruments each for measuring:
- (a) Mass
 - (b) Length
 - (c) Time
4. What is the weight of an object with a mass of 20 kg on Earth?
5. Explain why the weight of a person on Earth is different from the weight of the same person on the moon.

ASSIGNMENT

1. A student's mass is 60 kg:
 - (a) Calculate his weight on Earth.
 - (b) If gravity on the moon is 1.6 m/s^2 , what is his weight on the moon?
2. Describe three sources of error in measurement and how to avoid them.
 - Accuracy and proper use of instruments ensure reliable results in physics experiments.

WEEK 3 - POSITION, DISTANCE, AND DISPLACEMENT

DEFINITION OF POSITION, DISTANCE AND DISPLACEMENT

Position: This is the location of an object relative to a reference point (origin). Often described in terms of direction (left/right, north/south) and distance from the reference point.

Example:

- A student sitting 2 meters to the right of the board is said to be at **position +2m**.

Distance: This is the total length of the path covered by a moving object, regardless of direction. It is a scalar quantity (has magnitude but no direction). It is always positive and measured in metres (m).

Example: 1

If a person walks 4 meters forward and 4 meters back, the total distance is:

$$4\text{m} + 4\text{m} = 8\text{m}$$

Example: 2

If you walk 6 m east and 4 m west, the total distance is:

$$6\text{m} + 4\text{m} = 10\text{m}$$

Displacement: This is the shortest distance from the starting point to the final position of the object in a straight line and specific direction. It is a vector quantity (has both magnitude and direction). It can be positive, negative, or zero.

Example:

Using the same movement:

- From starting point to 4 m forward, and 4 m back = 0 m displacement (you returned to where you started).

COMPARISON TABLE – DISTANCE VS DISPLACEMENT

Feature	Distance	Displacement
Type of Quantity	Scalar	Vector
Direction	Not required	Required

Can be negative?	No	Yes (if opposite to reference direction)
Path or straight line	Depends on path	Shortest path
Symbol	D	s or Δx
Example (4m forth and 4m back)	8 m	0 m

GRAPHICAL AND PHYSICAL REPRESENTATION

Position, distance, and displacement can be graphically represented using graphs that plot position or distance against time. A position-time graph shows an object's location at different points in time, while a displacement-time graph shows the change in position (including direction) over time. Distance-time graphs, on the other hand, only show the total distance traveled, regardless of direction.

Position-Time Graph:

- The x-axis represents time, and the y-axis represents the object's position.
- A straight, horizontal line indicates the object is stationary.
- An upward-sloping line indicates the object is moving away from the origin (positive direction).
- A downward-sloping line indicates the object is moving towards the origin (negative direction).
- The slope of the line at any point represents the object's instantaneous speed.

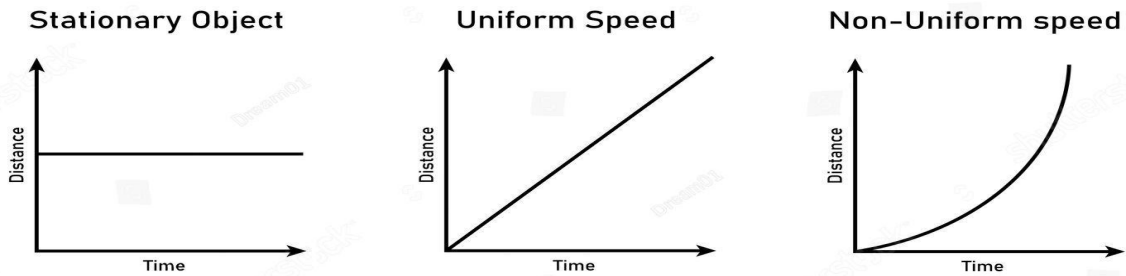
Displacement-Time Graph:

- Similar to a position-time graph, the x-axis represents time, and the y-axis represents displacement.
- Displacement is the change in position from the starting point, including direction.
- The slope of the line at any point represents the object's instantaneous velocity.

Distance-Time Graph:

- The x-axis represents time, and the y-axis represents the total distance traveled.
- A straight, horizontal line indicates the object is not moving.
- An upward-sloping line indicates the object is moving.
- The slope of the line at any point represents the object's speed.

Position-Time Graph



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Distance-time graphs do not show direction.

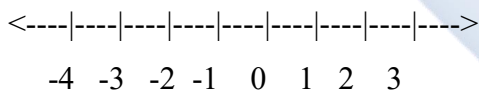
Draw a straight line on the floor or board representing a number line.

- Mark 0 as the **origin**.
- Mark +5 m to the right and -5 m to the left.

Demonstrate:

- Student walks from 0 to +5 m.
 - **Distance = 5 m, Displacement = +5 m**
- Student walks back to 0:
 - **Distance = 10 m, Displacement = 0 m**

Representation Using a Number Line



Numerical Examples

1. Example 1

A boy walks 6m east and then 3m west.

$$\text{Total distance} = 6 + 3 = 9\text{m}$$

$$\text{Displacement} = 6 - 3 = 3\text{m east}$$

2. Example 2

A girl walks 10m north, 10m south.

$$\text{Distance} = 10 + 10 = 20\text{m}$$

$$\text{Displacement} = 0\text{m (back to starting point)}$$

Real-Life Applications

- Measuring how far a vehicle or person has traveled.
- In athletics: 400 m on a round track may be a large **distance**, but **displacement** after one round is **zero**.
- In GPS tracking: Your displacement shows your current position from your starting location.

EVALUATION QUESTIONS (WAEC/NECO-BASED)

1. WAEC June 2007 (Paper 2, Q3a)

A motorist travels 100 km due North and then 60 km due East. Calculate the magnitude of his total displacement and the direction from the starting point.

(Take East as 0°)

Answer:

$$\text{Displacement} = \sqrt{(100^2 + 60^2)} = \sqrt{(10000 + 3600)} = \sqrt{13600} = \mathbf{116.6 \text{ km}}$$

$$\text{Direction, } \theta = \tan^{-1}(60/100) = \mathbf{31^\circ \text{ East of North}}$$

2. NECO June 2011 (Paper 1, Q5)

Which of the following quantities involves both magnitude and direction?

- A. Distance
- B. Speed
- C. Time
- D. Displacement

3. WAEC June 2015 (Paper 2, Q2c)

An athlete runs round a circular track of radius 70 m and returns to the starting point after 140 s. Calculate:

- i. *The distance covered*
- ii. *The displacement*

Answer:

- i. Distance = Circumference = $2\pi r = 2 \times 3.142 \times 70 = 439.88 \text{ m}$
- ii. Displacement = **0 m** (since start = end point)

4. WAEC June 2018 (Paper 1, Q10)

If a body moves from point A to point B and back to point A in 20 seconds, covering a total distance of 40 m, what is the displacement and average speed?

Answer:

Displacement = **0 m**

Average speed = Total distance / Total time = $40 / 20 = 2 \text{ m/s}$

5. NECO June 2019 (Paper 2, Q4b)

Explain the difference between distance and displacement, using a car that moves from point A to B and back to A as an example.

Answer:

Distance is the total length of the path covered, which is $A \rightarrow B \rightarrow A$ (say 60 m each way = 120 m).

Displacement is the change in position = 0 m, since the final position is the same as the initial position.

ASSIGNMENTS (WAEC/NECO PATTERN)

1. A boy walks 6 km north and then 8 km east. Calculate:
 - a. The total distance covered.

- b. The magnitude of the displacement.
- c. The direction of displacement.

2. Differentiate between scalar and vector quantities using the following terms:

- Distance
- Displacement
- Speed
- Velocity

3. A cyclist moves from point A to point B, a distance of 120 m in a straight line. He then returns halfway back towards A. Find:

- a. The total distance covered
- b. The magnitude of the displacement
- c. Why displacement is not equal to distance in this case.

4. A student walks around a rectangular field 40 m long and 30 m wide:

- a. What is the total distance covered in one complete lap?
- b. What is the displacement after one full lap?
- c. What will be the displacement if the student walks from one corner to the diagonally opposite corner?

5. State two differences between distance and displacement and give one real-life example to illustrate each.

WEEK 4 & 6: SCALAR AND VECTOR QUANTITIES

MEANING OF SCALAR QUANTITIES: These are Physical quantities that have only magnitude but no specific direction.

Common Scalar Quantities are:

- | | |
|----------------|------------|
| 1. Mass | 6. Energy |
| 2. Time | 7. Work |
| 3. Temperature | 8. Power |
| 4. Distance | 9. Density |
| 5. Speed | 10. Volume |

MEANING OF VECTOR QUANTITIES: These are Physical quantities that have both magnitude and direction.

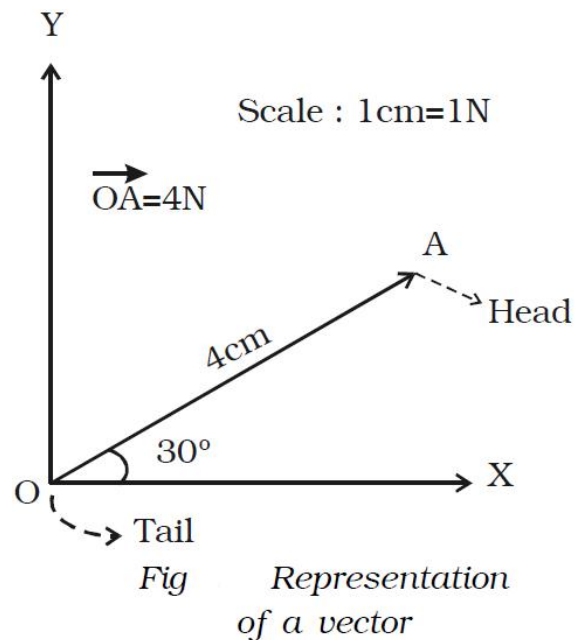
Common Vector Quantities are:

- | | |
|-----------------|-----------------------------|
| 1. Displacement | 5. Weight |
| 2. Velocity | 6. Electric field intensity |
| 3. Acceleration | 7. Momentum |
| 4. Force | 8. Magnetic field strength |

DIFFERENCES BETWEEN SCALARS AND VECTORS

Feature	Scalar Quantities	Vector Quantities
Definition	Magnitude only	Magnitude and direction
Addition Rule	Ordinary arithmetic	Parallelogram law
Representation	Number only	Arrow representation
Example	Temperature (30°C)	Force (20N East)
Behavior	Invariant under rotation	Changes with rotation

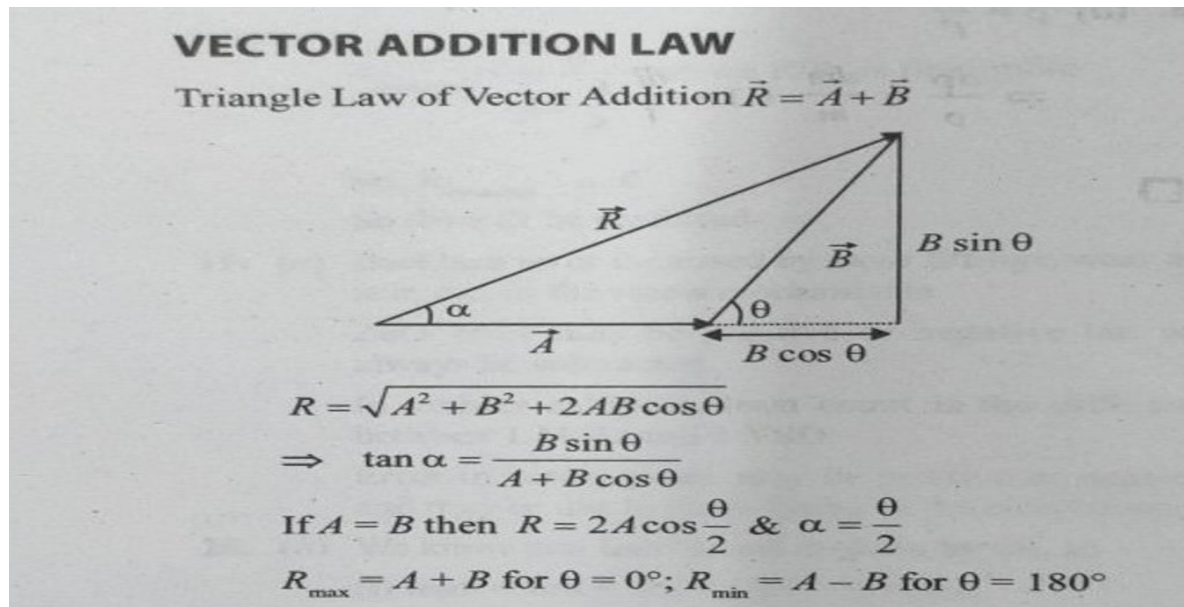
GRAPHICAL REPRESENTATION:



ADDITION VECTORS:

Vector addition takes into consideration not only the magnitude but the direction of the vectors. The methods of calculation also depend on whether the vectors are acting in a straight line or at an angle to each other.

1. two parallel forces acting in the same direction [$R = F_1 + F_2$]
2. two forces acting in opposite direction in a straight line [$R = F_1 - F_2$]
3. two forces acting perpendicular to each other (at right angle) [$F_1 \perp F_2$]. The resultant force R , is found using Pythagoras theorem. $R^2 = F_1^2 + F_2^2$
4. Angle between the two acting forces is ACUTE (0° and 90°) or Obtuse (90° and 180°)

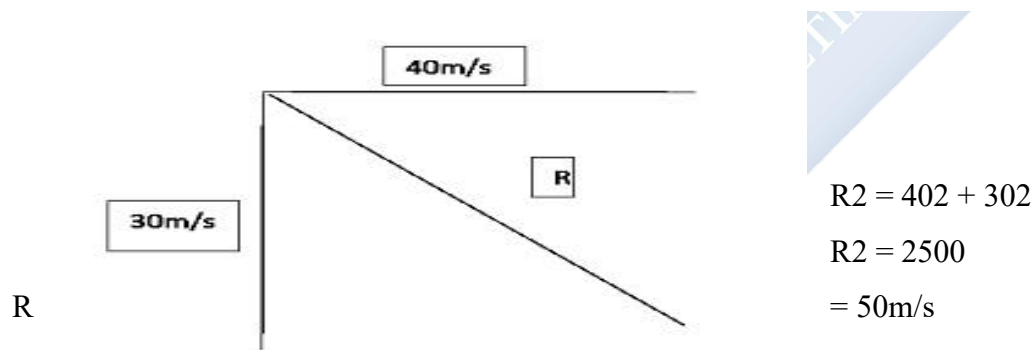


SIMPLE PROBLEMS INVOLVING VECTOR ADDITION

Example 1:

A boat travels due east at a speed 40ms^{-1} across a river flow due south at 30ms^{-1} . What is the resultant speed of the boat?

Solution



Example 2

An aircraft attempts to fly due north at 100kmh^{-1} . If the wind blows against it from east to west at 60kmh^{-1} , its resultant velocity is

Solution

Using Pythagoras theorem;

$$100^2 + 60^2 = (\text{resultant velocity i.e } RV)^2$$

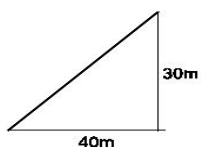
$$10000 + 3600 = RV^2; 13600 = RV^2$$

$$RV = 13600 = 116.7\text{kmh}^{-1} = 117\text{kmh}^{-1} \quad 8\text{nl} * \tan q / 100 \quad 60 = 0.6; \tan q = 0.6 = 30.96$$

$$310 = \text{N}310\text{W}; 117\text{kmh}^{-1}, \text{N}310\text{W}.$$

Example 3

In a school compound, a girl walks 40m due east from the laboratory to the staff room to submit her report. Then she turns and walks 30m due north to her classroom. Determine the magnitude of her displacement from the laboratory.



Solution

$$R^2 = 40^2 + 30^2$$

$$R^2 = 1600 + 900$$

$$R = \sqrt{2500} = 50\text{m}$$

EVALUATION:

1. Define:
 - (a) Scalar quantity
 - (b) Vector quantity
2. List any four scalar and four vector quantities.
3. Differentiate between scalar and vector quantities.
4. Represent a force of 6 N acting at 30° to the horizontal using an arrow.
5. A boy walks 3 km north, then 4 km east. What is:
 - (a) The total distance covered?
 - (b) The displacement? (Use Pythagoras' Theorem)

ASSIGNMENT:

Question 1 – Adapted from WAEC Physics, June 2014, Paper 2, Q1

(a) Define the following terms:

- (i) Scalar quantity
- (ii) Vector quantity

(b) List **three examples** each of scalar and vector quantities.

(c) A man walks 6 km due north, then 8 km due east.

- (i) What is the total distance covered?
- (ii) Calculate the magnitude of the resultant displacement.

(Take $\sqrt{100} = 10$)

Question 2 – NECO SSCE Physics, June 2018, Paper 2, Q2 (Adapted)

(a) Distinguish between scalar and vector quantities using any two physical examples.

(b) State **two major characteristics** that differentiate vector quantities from scalar quantities.

(c) A boy pushes a box with a force of 15 N at an angle of 30° to the horizontal.

Resolve the force into:

- (i) Horizontal component
- (ii) Vertical component

(Take $\cos 30^\circ = 0.866$, $\sin 30^\circ = 0.5$)

Question 3 – WAEC Physics, June 2016, Paper 2, Q3 (Adapted)

(a) Using a clearly labelled diagram, explain how two vectors can be added using the **parallelogram law of vector addition**.

(b) A car moves 12 km east and then 5 km south.

- (i) Represent the motion on a diagram.
- (ii) Determine the magnitude of the displacement.

(Leave your answer in two decimal places)

Question 4 – NECO Physics, June 2019, Paper 2, Q1 (Modified)

(a) State whether the following quantities are scalar or vector:

- (i) Speed
- (ii) Work
- (iii) Acceleration
- (iv) Force
- (v) Pressure

(b) Mention one physical situation where it is important to consider direction when dealing with motion.

Question 5 – WAEC Physics, June 2020, Paper 2, Q2 (Adapted)

(a) Define displacement and explain how it differs from distance.

(b) A cyclist travels 10 km due west and then returns 6 km due east.

- (i) What is the total distance travelled?
- (ii) What is the displacement of the cyclist?
- (iii) Is displacement a scalar or vector quantity? Justify your answer.

WEEK 8 - SPEED, VELOCITY AND ACCELERATION

DEFINITION OF SPEED: This is the distance traveled per unit time (scalar). The S.I. Unit of speed is in metre per second (m/s).

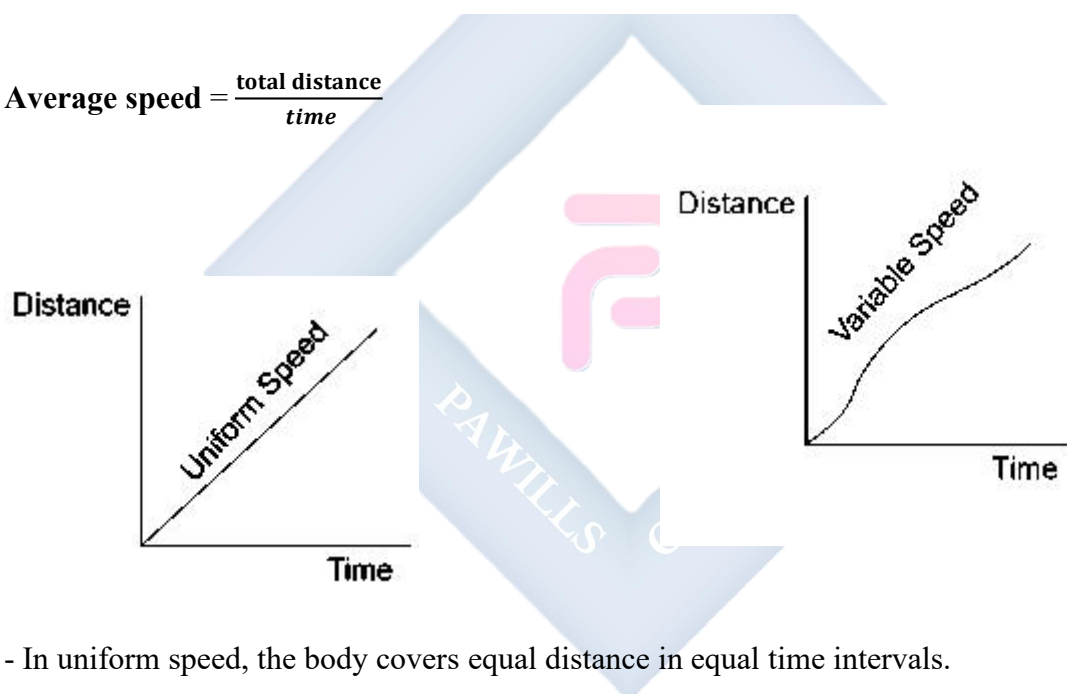
Formula:

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

Types of speed:

- **Uniform speed** – This is known as the constant speed (e.g., cruise control in a car).
- **Average speed** – This is the total distance covered over the time period. This can be expressed by the relation,

$$\text{Average speed} = \frac{\text{total distance}}{\text{time}}$$



- In uniform speed, the body covers equal distance in equal time intervals.

- **Instantaneous speed** – This is the Speed at a particular moment.

DEFINITION OF VELOCITY: Velocity is the rate of change of displacement with time (vector). The S.I. Units is in m/s.

Formula:

$$\text{Velocity} = \frac{\text{Displacement}}{\text{Time}}$$

Uniform velocity: The velocity of a body is said to be uniform, if it is displaced

Equally in equal time no matter how small the time interval may be.

DEFINITION OF ACCELERATION: Acceleration is the rate of change of velocity with time.

The S. I. Units is in m/s^2

Formula:

$$\text{Acceleration} = \frac{\text{Change in Velocity}}{\text{Time taken}}, \quad a = \frac{v-u}{t},$$

Where u = initial velocity, v = final velocity and t = time taken

Types Of Acceleration:

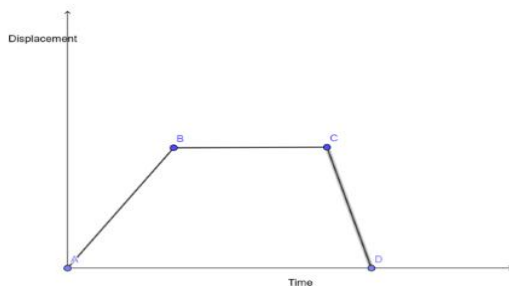
- **Positive acceleration** – Velocity increases (e.g., speeding up).
- **Negative acceleration (Deceleration)** – Velocity decreases (e.g., braking).
- **Uniform acceleration** – Constant change in velocity (e.g., free-fall under gravity).

DIFFERENCES BETWEEN SPEED AND VELOCITY

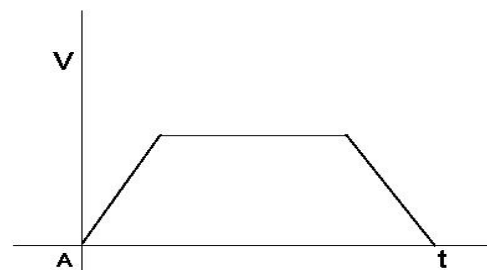
Speed	Velocity
Scalar quantity (magnitude only)	Vector quantity (magnitude + direction)
Always positive	Can be positive or negative (direction matters)
Depends on distance	Depends on displacement
Example: 50 km/h	Example: 50 km/h East

GRAPHS OF MOTION

Displacement-Time Graph



Velocity-Time Graph



CALCULATIONS ON SPEED, VELOCITY, AND ACCELERATION USING APPROPRIATE FORMULAS

Example 1:

A cyclist covers 150 meters in 30 seconds. What is the speed?

Solution

By applying the formula;

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{150}{30}$$

$$= 5 \text{ m/s}$$

Example 2:

A runner moves 200m East in 40s. What is the velocity?

Solution

$$\text{Velocity} = \frac{200}{40} = 5 \text{ m/s}$$

Example 3:

A car accelerates from 10 m/s to 30 m/s in 5 seconds. Find acceleration.

Solution

$$\text{Acceleration} = \frac{v-u}{t} = \frac{30-10}{5} = 4 \text{ m/s}^2$$

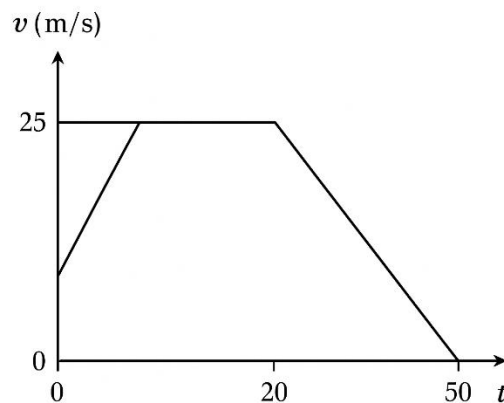
DETERMINATION OF ACCELERATION, RETARDATION AND DISTANCE FROM VELOCITY-TIME GRAPH

Example 1:

A car starts from rest and accelerates uniformly for 20 s to attain a speed of 25 ms⁻². It maintains this speed for 30 s before decelerating uniformly to rest. The total time for the journey is 60 s

- (a) Sketch a velocity - time graph for the motion.
- (b) Use the graph to determine the:
total distance travelled by the car
(Leaving the answer in whole number)

Solution



(a) Velocity–time sketch

Velocity starts at 0, rises linearly to 25 m/s in 20 s, stays constant at 25 m/s for 30 s (from $t = 20$ s to $t = 50$ s), then falls linearly to 0 in the last 10 s ($t = 50 \rightarrow 60$ s).

(At $t = 0$ $v = 0$; at $t = 20$ $v = 25$; horizontal line $v = 25$ from $t = 20$ to $t = 50$; linear drop to $v = 0$ at $t = 60$.)

(b) Total distance — area under the v–t graph

Break into three areas:

1. Acceleration phase (triangle): area = $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 20 \text{ s} \times 25 \text{ m/s} = 250 \text{ m}$.
2. Constant speed phase (rectangle): area = $30 \text{ s} \times 25 \text{ m/s} = 750 \text{ m}$.
3. Deceleration phase (triangle): area = $\frac{1}{2} \times 10 \text{ s} \times 25 \text{ m/s} = 125 \text{ m}$.

Total distance = $250 + 750 + 125 = 1125 \text{ m}$.

(Whole-number answer: **1125 m**.)

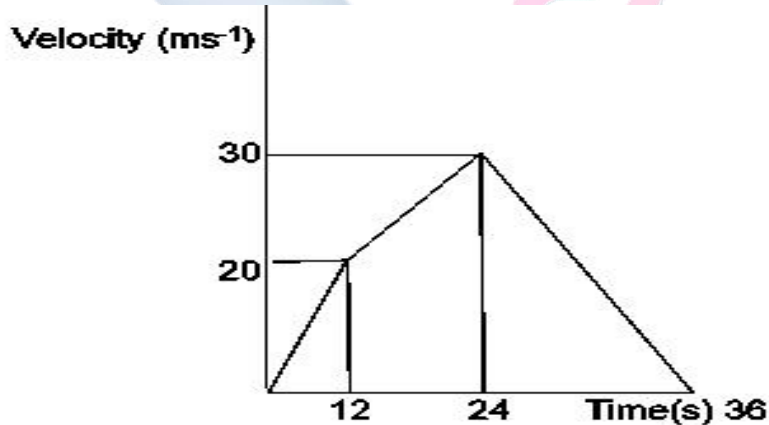
Example 2:

A body start from rest and travels distance of 120, 300 and 180m in successive equal time intervals of 12s.

During each interval the body is uniformly accelerated

Sketch a velocity-time graph for the motion.

Solution



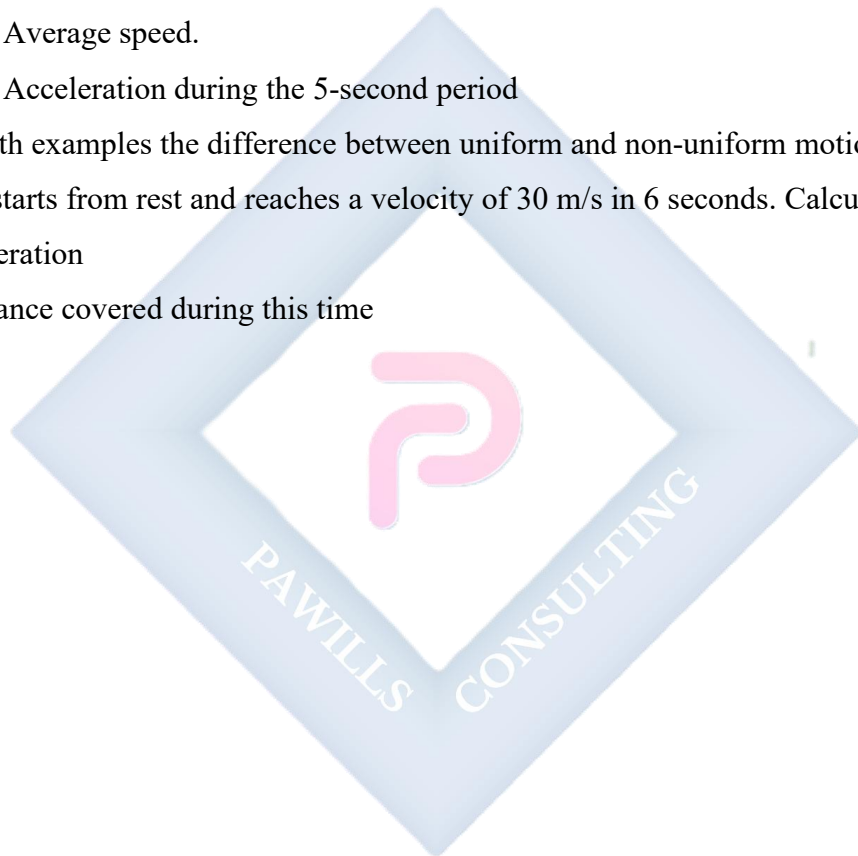
EVALUATION

1. What is the difference between speed and velocity?
2. Can an object have zero speed but non-zero acceleration? Explain.
3. A train travels 500 km in 5 hours. What is its average speed?
4. A car decelerates from 25 m/s to 10 m/s in 3 seconds. Find acceleration.

5. Sketch a velocity-time graph for:
- Uniform acceleration
 - Constant velocity
 - Deceleration

ASSIGNMENT:

- A car moves at 15 m/s for 10 seconds, then accelerates to 25 m/s in 5 seconds. Calculate:
 - Total distance covered.
 - Average speed.
 - Acceleration during the 5-second period
- Explain with examples the difference between uniform and non-uniform motion.
- A vehicle starts from rest and reaches a velocity of 30 m/s in 6 seconds. Calculate:
 - Its acceleration
 - The distance covered during this time



WEEK 9 & 10 - MOTION

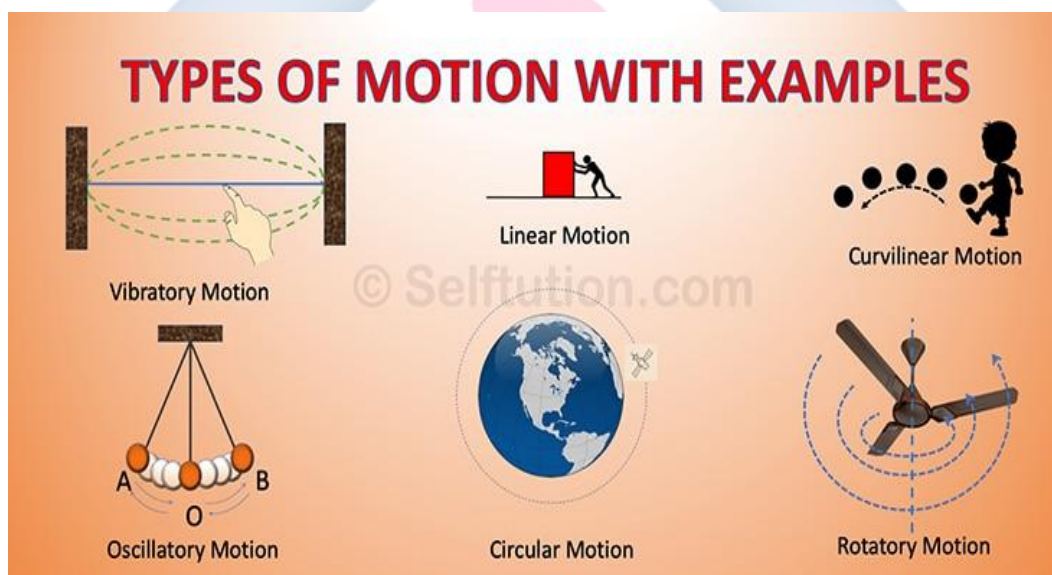
CONCEPT OF MOTION

Motion is the change in position of a body with respect to time and a reference point.

- **Rest:** A body is said to be at rest if it does not change its position with time.
- **Motion:** A body is in motion if it changes position with time.

TYPES OF MOTION

- **Translational Motion:** Movement in a straight or curved path.
- **Rotational Motion:** When a body spins about an axis.
- **Oscillatory Motion:** Back-and-forth motion about a fixed point (e.g., pendulum).
- **Random Motion:** Irregular, unpredictable movement (e.g., gas molecules).
- **Linear Motion:** A type of translational motion in a straight line.



CAUSES OF MOTION

2. FORCE

Force is a **push or pull** that tends to change the **state of rest or motion** of a body, or its **shape**.

It is a **vector quantity**, meaning it has both **magnitude and direction**.

SI unit of force: Newton (N)

Formula:

$$F = \text{mass} \times \text{acceleration}$$

$$= m \times a$$

$$= ma$$

Where:

F = force (N), m = mass (kg), a = acceleration (m/s^2)

Effects of Force: Force can;

- Start or stop motion.
- Change the direction of an object.
- Change the speed of an object.
- Change the shape or size of an object.

Types of Force

a. Contact Forces:

- Applied force
- Frictional force
- Tension force
- Normal force
- Air resistance

b. Non-contact (Field) Forces:

- Gravitational force
- Magnetic force
- Electrostatic force

3. **Friction:** This is a **force that opposes relative motion** or the tendency of such motion between two surfaces in contact.

Types of Friction

Type	Description
Static Friction	The force that must be overcome to start the motion of a stationary object.
Kinetic (Dynamic) Friction	The friction acting when an object is already in motion.

Note:

- Static friction is **greater** than kinetic friction.
- Frictional force is proportional to the **normal reaction**:

$$F_{\text{friction}} = \mu N$$

Where: μ = coefficient of friction (dimensionless), N = normal force (N)

Advantages of Friction

- Enables walking and driving without slipping.
- Allows brakes to work in vehicles.
- Useful in gripping and holding tools.
- Enables machines like belts and pulleys to function.

Disadvantages of Friction

- Causes wear and tear in machines.
- Leads to energy loss as heat.
- Reduces efficiency in moving parts.
- Makes movement difficult in some machinery.

Methods of Reducing Friction

- Lubrication (e.g., oil, grease).

- Polishing surfaces.
- Using ball bearings.
- Streamlining bodies (especially in vehicles and aircraft).

Methods of Increasing Friction

- Using rough surfaces (e.g., sandpaper).
- Increasing normal force (weight).
- Using treads on tyres.
- Applying rubber soles for grip.

Applications of Force and Friction

- Braking systems in vehicles
- Movement of conveyor belts
- Operation of engines
- Design of footwear
- Climbing (grip between hands and surfaces)
- Road and tyre design

A box of mass 5 kg is pulled on a horizontal floor with a force of 30 N. If the coefficient of kinetic friction between the box and the floor is 0.2, calculate the acceleration of the box.

Solution:

Given:

$$m = 5\text{kg}, F = 30\text{N}, \mu = 0.2, g = 9.8 \text{ m/s}^2$$

$$\text{Normal reaction } N = mg = 5 \times 9.8 = 49 \text{ N}$$

$$\text{Frictional force } F_r = \mu N = 0.2 \times 49 = 9.8\text{N}$$

$$\text{Net force } F_{\text{net}} = 30 - 9.8 = 20.2\text{N}$$

$$\text{Acceleration } a = \frac{F_{\text{net}}}{m} = \frac{20.2}{5} = 4.04\text{m/s}^2$$

EQUATIONS OF MOTION.

If an object travelling at uniform acceleration, a , increases its velocity from a value u , to a final value v , in a time t , its average velocity = $\frac{v+u}{2}$,

Distance (s) travelled by the object in this time is given by,

$$S = \text{average velocity} \times t$$

$$S = \frac{(v+u)t}{2}, \text{-----(i)}$$

$$\text{From the definition of acceleration, } a = \frac{v-u}{t}$$

$$\text{The final velocity attained is given by, } v = u + at \text{-----(ii)}$$

By eliminating v in eqn (i)

$$\text{Distance (s)} = ut + \frac{1}{2}at^2 \text{-----(iii)}$$

By eliminating t another useful equation can be obtained from eqn (ii)

$$t = \frac{v-u}{a} \text{ (substituting this into equation 1)}$$

$$s = \left(\frac{v+u}{2} \right) \left(\frac{v-u}{a} \right)$$

$$\text{hence, } v^2 = u^2 + 2as \text{-----(iv)}$$

These equations are valid under uniform acceleration:

$$1. \quad V = u + at$$

$$2. \quad S = ut + \frac{1}{2}at^2$$

$$3. \quad V^2 = u^2 + 2as$$

Where: s = displacement

u = initial velocity

v = final velocity

a = acceleration

t = time

PROBLEMS INVOLVING EQUATIONS OF MOTION

Example 1:

A car accelerates uniformly from rest at 2 m/s^2 for 10 seconds. Calculate:

- (a) The final velocity
- (b) The distance covered

Solution:

- (a) By applying the formula,

$$\begin{aligned}v &= u + at \\&= 0 + 2 \times 10 \\&= 20 \text{ m/s}\end{aligned}$$

- (b) $s = ut + \frac{1}{2}at^2$
- $$\begin{aligned}&= 0 + 2 \times 10 \times 10 \\&= 100 \text{ m}\end{aligned}$$

Example 2:

An object moves with an initial velocity of 5 m/s , and accelerates uniformly to a velocity of 25 m/s over a distance of 60 m . Find the acceleration.

Solution:

Using the formula, $v^2 = u^2 + 2as$

$$\begin{aligned}25^2 &= 5^2 + 2a(60) \\ \Rightarrow 625 &= 25 + 120a \\ \Rightarrow 600 &= 120a \\ \Rightarrow a &= 5 \text{ m/s}^2\end{aligned}$$

Example 3:

A body accelerates uniformly from rest at 2ms^{-2} . Calculate its velocity when it has travelled a distance of 9m .

Solution

$$a = 2\text{m/s}^2, u = 0, s = 9\text{m}, v = ?$$

Using $v^2 = u^2 + 2as$

$$v^2 = 0^2 + 2 \times 2 \times 9$$

$$v^2 = 36$$

$$v = \sqrt{36} = 6\text{m/s}$$

GRAVITATIONAL ACCELERATION:

When an object is thrown upwards or released from a height, its motion is governed or affected by gravity. Equation of motion for gravitational acceleration is derived from the equations of motion for rectilinear acceleration:

$$V = u - gt \text{----- (1)}$$

$$h = ut - \frac{1}{2}gt^2 \text{----- (2)}$$

$$v^2 = u^2 - 2gs \text{----- (3)}$$

Example:

A particle is projected horizontally at 10ms^{-1} from the top of a tower 20m high. Calculate the horizontal distance travelled by the particle when it hits the level ground. [$g = 10\text{ms}^{-2}$]

Solution

Parameters given; Initial velocity, $U = 0\text{ms}^{-1}$, height, $S = 20\text{m}$

Recall that:

$$S = ut + \frac{1}{2}gt^2$$

$$20 = 0 \times t + \frac{1}{2} \times 10 \times t^2$$

$$20 = 0 + 5t^2$$

$$t^2 = 20/5$$

$$t = \sqrt{4}$$

$$t = 2\text{sec}$$

$$t = 2\text{sec}, u = 20\text{m/s}$$

EVALUATION

1. A body moves with an initial velocity of 12 m/s and accelerates uniformly at 3 m/s² for 4 seconds. Find:
 - a. The final velocity
 - b. The total distance covered
2. A body starts from rest and travels distance of 120, 300 and 180m in successive equal time intervals of 12s.

During each interval the body is uniformly accelerated.

Calculate the velocity of the body at the end of **each** successive time interval.

ASSIGNMENT

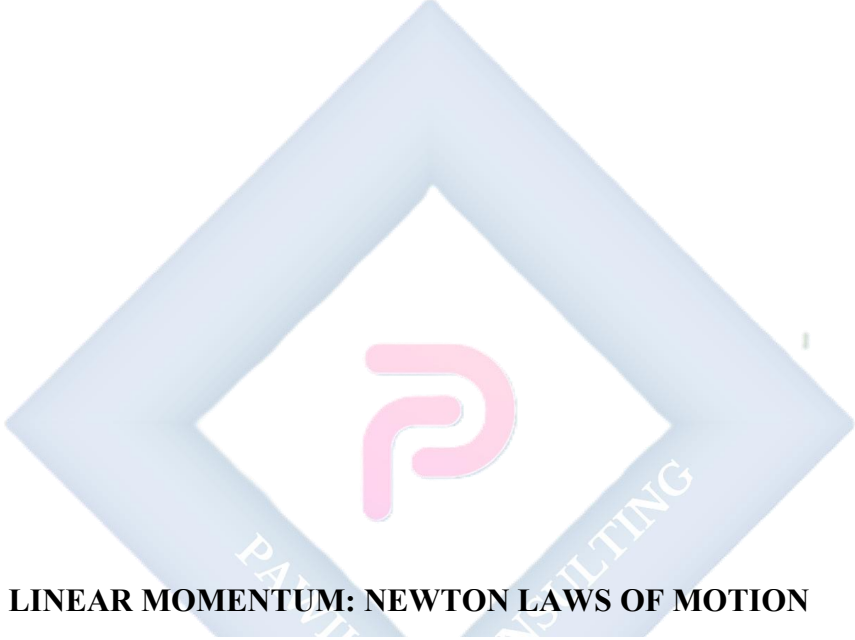
4. A car takes off from rest and covers a distance of 80m on a straight road in 10s. calculate the magnitude of its acceleration
5. A body starts with an initial velocity of 20 ms⁻¹ and moves with a uniform acceleration, a, for 15 seconds. Which of the following equations represents the distance, S, travelled?

SS1 SECOND TERM LESSON CONTENTS ON PHYSICS

Scheme of work

1. Linear Momentum: Newton Laws of Motion
2. Work, Energy and Power
3. Mechanical Energy: Machines
4. Heat Energy (i) Temperature and its Measurement
5. Heat Energy Measurements
6. Particulate Nature of Matter





WEEK ONE - LINEAR MOMENTUM: NEWTON LAWS OF MOTION

DEFINITION OF LINEAR MOMENTUM

This simply refers to as the product of mass and velocity ($p = mv$). The SI unit is in kgm/s.

Mathematically, momentum = mass (m) x velocity(v)

Example 1:

A 5kg object moves at 10m/s. Calculate its momentum.

Solution:

$$p = mv = 5 \times 10 = 50\text{kgm/s}$$

- Impulse is defined as the change in momentum produce by the force.

$$\text{Force} = \frac{\text{change in momentum}}{\text{Time}}$$

$$I = Ft = Mv - Mu$$

I = Impulse, F = force, t = time, M = mass of body, V = final velocity, U = Initial velocity

Example 2:

A tractor of mass $5.0 \times 10^3 \text{ kg}$ is used to tow a car mass $2.5 \times 10^3 \text{ kg}$. The tractor moved with a speed of 3.0 ms^{-1} just before the towing rope becomes taut. Calculate the; impulse in the rope when it jerks the car into motion

Solution

$$\begin{aligned} Ft &= M_1 V_1 - M_2 V_2 \\ &\Rightarrow 5.0 \times 10^3 \times 3 - 2.5 \times 10^3 \times 2 \\ &\Rightarrow (15 - 5) \times 10^3 \\ &\Rightarrow 10 \times 10^3 \text{ Ns} \end{aligned}$$

LAW OF CONSERVATION OF LINEAR MOMENTUM:

The law states that in the system of colliding objects the total momentum is conserved, provided there is no external force acting on it.

Total momentum before collision = Total momentum after collision (for isolated system)

$$M_1 u_1 + M_2 u_2 = M_1 v_1 + M_2 v_2$$

Where:

u = initial velocities

v = final velocities

m = masses

Types of Collisions:

1. Elastic collisions

2. Inelastic collisions
3. Perfectly inelastic collision

Example:

A 2kg object accelerates from 5m/s to 15m/s in 4s. Calculate:

- a) Change in momentum
- b) Applied force

Solution:

- a) $\Delta p = m(v-u) = 2(15-5) = 20\text{kgm/s}$
- b) $F = \Delta p / \Delta t = 20/4 = 5\text{N}$

NEWTON'S LAWS OF MOTION

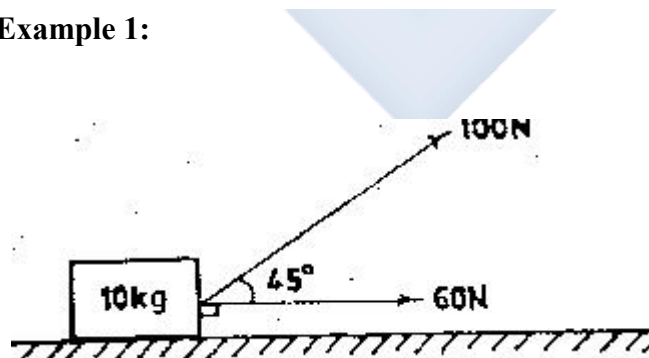
First Law (Law of Inertia): An object remains at rest or in uniform motion unless acted upon by external force.

Inertia: This is the reluctance of a body at rest to be in motion and the reluctance of a body in uniform motion to change its velocity.

Second Law ($F=ma$): states that the force applied on a body is directly proportional to its time rate of change in momentum and takes place in the direction of the force.

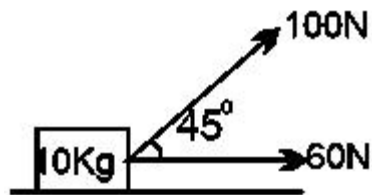
Third Law (Action-Reaction): For every action, there's equal and opposite reaction.

Example 1:



Two forces 100N and 60N act on an object of mass 10Kg as illustrated in the diagram above. Neglecting friction, calculate the acceleration of the object.

Solution

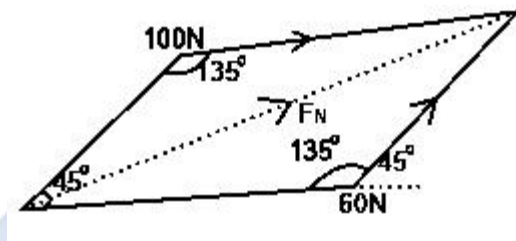


Resolving 100N into horizontal and vertical

$$100\sin 45 = 70.71; 100\cos 45 = 70.71$$

$$\text{vertical forces} = 100 - 70.71 = 29.29$$

$$\text{Horizontal force} = 70.71 + 60 = 130.71$$



Resultant of vertical and horizontal forces using pythagoras theorem

$$R = 133.9\text{N from } F = ma$$

$$F = 133.9$$

$$a = \frac{F}{m} = \frac{133.9}{10} = 13.4 \text{ m/s}^2$$

Example: 2

An external force of magnitude 100 N acts on a particle of mass 0.15kg for 0.03s. Calculate the change in the speed of the particle.

Solution

Given:

Force, $F = 100\text{N}$, mass, $m = 0.15\text{kg}$, Time, $t = 0.03\text{s}$; change in speed, $\Delta v = ?$

Recall:

$$F = \frac{m\Delta v}{t} \quad Ft = m\Delta v;$$

$$\Delta v = \frac{100 \times 0.03}{0.15} = \frac{3}{0.15} = 20 \text{ms}^{-1}$$

REAL-LIFE APPLICATIONS

- Seat belts in cars (First Law)
- Rocket propulsion (Third Law)
- Braking systems (Second Law)
- Sports (e.g., kicking a ball)

EVALUATION

1. State Newton's three laws of motion
2. A 1200kg car moving at 25m/s brakes to stop in 5s. Calculate:
 - a) Initial momentum
 - b) Average braking force
3. Explain how Newton's third law applies to:
 - a) Walking
 - b) Rocket launch
4. Solve: A 0.5kg ball hits a wall at 10m/s and rebounds at 8m/s. If contact lasts 0.2s, calculate the average force.

ASSIGNMENTS

1. A body of mass 5kg moving with a velocity of 10ms⁻² collides with a stationary body of mass 6kg. If the two bodies stick together and move in the same direction after the collision, calculate their common velocity.

2. A net force of 15 N acts upon a body of mass 3kg for 5s. Calculate the change in the speed of the body.
3. A trolley of mass 4 kg moving on a smooth horizontal platform with a speed of 1.0 ms⁻¹ collides perfectly with a stationary trolley of the same mass on the same platform. Calculate the total momentum of the two trolleys immediately after the collision

WEEK TWO - WORK, ENERGY AND POWER

DEFINITION OF WORK: Work is said to be done when a force moves an object through a distance in the direction of the force.

Formula for Work Done

$$W = \text{Force} \times \text{distance} \cos\theta$$

$$W = F \times d \times \cos\theta$$

Where:

W = Work done (Joules, J)

F = Force applied (Newtons, N)

d = Displacement (meters, m)

θ = Angle between the force and the direction of motion

Conditions for Work to Be Done

- A force must be applied.
- There must be displacement.
- The displacement must have a component in the direction of the force.

Types of Work

- **Positive Work:** Force and displacement in same direction.
- **Negative Work:** Force opposes displacement (e.g., friction).
- **Zero Work:** No displacement or force perpendicular to displacement (e.g., carrying a load horizontally).

DEFINITION OF ENERGY: Energy is the capacity to do work. The S. I. Units of work is in Joule (J).

FORMS OF ENERGY

- Mechanical Energy: Kinetic and Potential
- Chemical Energy
- Thermal (Heat) Energy
- Electrical Energy
- Sound Energy
- Light Energy
- Nuclear Energy

Mechanical Energy

- Kinetic Energy (K.E): Energy of a moving body

$$K.E = \frac{1}{2}mv^2$$

- Potential Energy (P.E): Energy of a body due to its position or height

$$P.E = mgh$$

PRINCIPLE OF CONSERVATION OF ENERGY

Energy can neither be created nor destroyed but can only be transformed from one form to another. The total energy of an isolated system remains constant.

Example: A ball thrown upwards converts kinetic energy to potential energy. On falling back, potential energy is converted back to kinetic energy.

Worked Examples

Example 1:

A force of 20 N moves an object through a distance of 5 m. Calculate the work done.

Solution:

$$W = F \times d = 20 \times 5 = 100J$$

Example 2:

Find the kinetic energy of a 2 kg object moving at 3 m/s.

Solution:

$$\begin{aligned} \text{K.E} &= \frac{1}{2}mv^2 \\ &= \frac{1}{2} \times 2 \times 32 \\ &= 1 \times 9 = 9 \text{ J} \end{aligned}$$

DEFINITION OF POWER: Power is the rate at which work is done or energy is transferred.

Formula;

$$P = W \times t$$

Where:

P = Power (Watts, W)

W = Work done (J)

t = Time taken (s)

Unit of Power

- **Watt (W):** 1 W = 1 J/s
- **Kilowatt (kW):** 1 kW = 1000 W
- **Horsepower (hp):** 1 hp = 746 W

Example

A machine does 3000 J of work in 60 seconds. Find its power.

Solution:

$$P = W \times t = 3000 \times 60 = 50 \text{ W}$$

More examples from WAEC/NECO past questions

QUESTION 1 – WAEC Physics, June 2017, Paper 2, Q3 (Adapted)

- (a) Define the term *work* and state the SI unit.
- (b) A force of 50 N is used to push a cart through a distance of 20 m. Calculate the work done.
- (c) Under what condition is work said to be done?
- (d) Is any work done when a boy pushes hard against a wall but it does not move? Explain.

Answer:

(a)

Work is said to be done when a force is applied on an object and the object moves in the direction of the force.

SI Unit: Joule (J)

(b)

$$\text{Work done} = \text{Force} \times \text{Distance} = 50 \times 20 = 1000\text{J}$$

(c)

Work is said to be done only when there is displacement of an object in the direction of the applied force.

(d)

No, **no work is done** because the wall does not move. Since displacement is zero, work done is zero.

QUESTION 2 – NECO Physics, June 2019, Paper 2, Q4 (Adapted)

(a) Define:

(i) *Kinetic energy*

(ii) *Potential energy*

(b) A ball of mass 0.5 kg is kicked with a speed of 10 m/s. Calculate its kinetic energy.

(c) A body of mass 4 kg is lifted to a height of 5 m above the ground. Calculate its potential energy. (Take $g=10 \text{ m/s}^2$)

Answer:

(a)

(i) *Kinetic energy* is the energy possessed by a body due to its motion.

(ii) *Potential energy* is the energy possessed by a body due to its position or state.

(b)

$$\text{K.E.} = \frac{1}{2}mv^2$$

$$= \frac{1}{2} \times 0.5 \times 10^2 = 0.25 \times 100 = 25 \text{ J}$$

(c)

$$\text{P.E.} = mgh = 4 \times 10 \times 5 = 200 \text{ J}$$

QUESTION 3 – WAEC Physics, June 2015, Paper 2, Q2 (Adapted)

(a) Define *power* and state its SI unit.

(b) A machine does 1500 J of work in 25 seconds. Calculate its power output.

(c) A man of mass 70 kg runs up a flight of stairs 5 m high in 7 seconds. Calculate the power developed by the man.

(Take $g=10 \text{ m/s}^2$)

Answer:

(a)

Power is the rate at which work is done or energy is transferred.

SI Unit: Watt (W)

(b)

$$\text{Power} = \frac{\text{Work done}}{\text{Time}} = \frac{1500}{25} = 60 \text{ W}$$

(c)

Work done (or energy used) = $mgh = 70 \times 10 \times 5 = 3500\text{J}$

$$\text{Power} = \frac{3500}{7} = 500 \text{ W}$$

Evaluation Questions

1. Define work and state its SI unit.
2. What is the work done in moving a 10 N object through 8 m in the direction of the force?
3. List and explain three forms of energy.
4. A 4 kg object is raised to a height of 10 m. Calculate its potential energy.
5. Define efficiency and calculate it for a machine that uses 500 J to do 400 J of useful work.

Assignment

1. A car of mass 1000 kg moves at 10 m/s. Calculate its kinetic energy.
2. A boy exerts a force of 50 N to push a wheelbarrow 10 m. Find the work done.
3. If a pump transfers 6000 J of water in 30 s, what is its power?

WEEK 3 & 4: MECHANICAL ENERGY: MACHINES

INTRODUCTION TO MACHINES

A **machine** is any device that allows us to do work more easily by either:

- Multiplying the force we apply (making a small effort lift a large load),
- Changing the direction of the applied force, or
- Increasing the speed or distance moved.

Examples: Lever, Pulley, Inclined Plane, Screw, Wedge, Wheel and Axle.

BASIC FUNCTIONS OF A MACHINE

- To multiply force (**make work easier**)
- To change the direction of force (**e.g., fixed pulley**)
- To multiply distance or speed (**e.g., bicycle gears**)
- To transfer force from one point to another

TYPES OF SIMPLE MACHINES

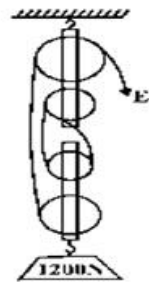
A. Lever: This is a rigid bar that pivots around a point (fulcrum). Divided into:

- **First Class Lever:** Fulcrum between effort and load (e.g., see-saw, scissors)
- **Second Class Lever:** Load between effort and fulcrum (e.g., wheelbarrow)
- **Third Class Lever:** Effort between fulcrum and load (e.g., tongs, fishing rod)

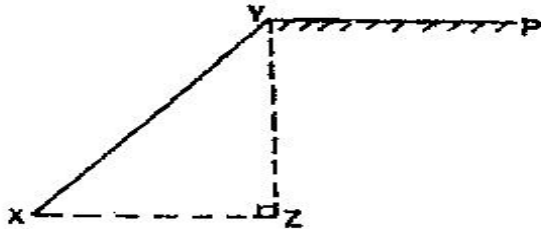
B. Pulley: This is a wheel on an axle designed to support movement and change of direction of a cable or belt.

- **Fixed Pulley** – changes direction only.
- **Movable Pulley** – multiplies effort.
- **Pulley System** – combines both.

V. R = number of pulleys in the system



C. Inclined Plane: This is a flat surface set at an angle to lift objects using less effort (e.g., ramp, staircase).



$$V.R = \frac{1}{\sin \theta}$$

D. Wheel and Axle: This is a circular frame (wheel) rotating with a central rod (axle) (e.g., steering wheel, doorknob).

$$V.R = \frac{R}{R}$$

E. Screw Jack: This is an inclined plane wrapped around a cylinder (e.g., screw jack, drill bit).

$$V.R = \frac{2\pi}{p}$$

F. Wedge: Two inclined planes joined together to split or cut (e.g., axe, chisel, knife).

$$V.R = \frac{R}{r}$$

IMPORTANT TERMINOLOGIES IN MACHINES

Term	Definition
Effort (E)	The force applied to operate a machine

Term	Definition
Load (L)	The resistance or weight to be moved
Mechanical Advantage (MA)	The ratio of load to effort.
Velocity Ratio (VR)	The ratio of the distance moved by effort to distance moved by load.
Efficiency (η)	The ratio of useful work output to total work input.

FORMULAE AND DERIVATIONS

Mechanical Advantage (MA):

$$MA = \frac{\text{load}}{\text{effort}}$$

$$\text{Velocity Ratio (VR)} = \frac{\text{Distance moved by Effort}}{\text{Distance moved by Load}}$$

$$\text{Efficiency} = \frac{\text{Work output}}{\text{Work input}} \times \frac{100}{1}$$

$$\text{Work output} = \text{Load} \times \text{Distance moved by load}$$

$$\text{Work input} = \text{Effort} \times \text{Distance moved by effort}$$

$$\text{Efficiency} = \frac{\text{Load} \times \text{Distance moved by Load}}{\text{Effort} \times \text{Distance moved by Effort}}$$

$$\text{Efficiency} = M.A \times \frac{1}{VR} \times 100\%$$

V.R

PROBLEMS INVOLVING MA, VR, AND EFFICIENCY

Example 1:

A block and tackle system has six pulleys. A force of 50N applied to it lifts a load of weight W. If the efficiency of the system is 40%, calculate W.

Solution

Given:

The velocity ratio, V.R = Number of pulleys = 6

The Effort, E = 50N

The Load, L = W

Efficiency = 40%

$$\text{Recall: Efficiency} = \left(\frac{L}{E} \right) \times \frac{1}{\text{V.R}}$$

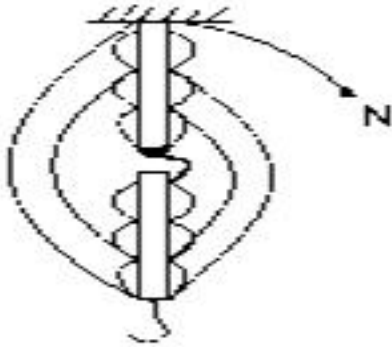
$$\therefore \frac{40}{100} = \frac{W}{50} \times \frac{1}{6}$$

$$\frac{4}{10} = \frac{W}{300}$$

$$W = 120\text{N}$$

Example 2:

The diagram illustrates a block and tackle pulley system in which an effort (N) supports a load of 100.0N. If the efficiency of the machine is 75%, calculate the value of N.

**Solution**

V.R = Number of pulleys = 6

M.A = ?

Efficiency Formula

$$E = \frac{M.A}{V.R} \times 100$$

Given:

$$E = 75\% = 0.75$$

Mechanical Advantage Formula

$$M.A = \frac{\text{Load}}{\text{Effort}}$$

Given:

- Efficiency = 0.75
- Effort = 6 N

Step 1: Find Mechanical Advantage (M.A)

$$0.75 = \frac{M.A}{6}$$

$$M.A = 0.75 \times 6 = 4.5$$

Step 2: Find Effort when Load = 100 N

$$M.A = \frac{100}{\text{Effort}}$$

$$4.5 = \frac{100}{\text{Effort}}$$

$$\text{Effort} = \frac{100}{4.5} = 22.2 \text{ N}$$

Example 3:

A simple machine with an efficiency of 75% lifts a load of 5000 N when a force of 500N is applied to it. Calculate the velocity ratio of the machine

Solution

A simple machine has:

- Load = 5000 N
- Effort = 500 N
- Efficiency = 0.75

Step 1: Mechanical Advantage

$$M.A = \frac{5000}{500} = 10$$

Step 2: Find Velocity Ratio (V.R)

$$0.75 = \frac{10}{V.R}$$

$$V.R = \frac{10}{0.75} = 13.3$$

APPLICATIONS OF MACHINES IN DAILY LIFE

- **Lever:** Crowbar, scissors, pliers
- **Pulley:** Lifting cranes, flag hoist
- **Inclined Plane:** Ramp for loading goods
- **Wheel and Axle:** Cars, rolling chairs
- **Wedge:** Axe, chisel
- **Screw:** Jack, bottle cap, car screw

EVALUATION QUESTIONS (WAEC/NECO-BASED)

1. WAEC June 2017 (Paper 2, Q4a)

A machine of efficiency 75% is used to lift a load of 600 N through a vertical height of 2 m.

Calculate:

- The work done on the load
- The work input to the machine

Answer:

i. Work output = Load $\times$ height = $600 \times 2 = 1200 \text{ J}$

ii. Work input = Work output / Efficiency
 $= 1200 / 0.75 = 1600 \text{ J}$

2. WAEC June 2006 (Paper 2, Q1c)

The velocity ratio of a machine is 5. If the machine is 80% efficient, calculate the mechanical advantage.

Answer:

$$\text{Efficiency} = (\text{MA} / \text{VR}) \times 100$$

$$80 = (\text{MA} / 5) \times 100 \rightarrow \text{MA} = (80 \times 5) / 100 = 4$$

3. NECO June 2015 (Paper 2, Q3b)

An inclined plane 5 m long is used to raise a load through a vertical height of 1.25 m. If a force of 300 N is applied to push the load up the plane, and the load has a weight of 1000 N, calculate:

- The velocity ratio
- The mechanical advantage
- The efficiency of the machine

Answer:

i. $VR = \text{Length} / \text{Height} = 5 / 1.25 = 4$

ii. $MA = \text{Load} / \text{Effort} = 1000 / 300 = 3.33$

iii. $\text{Efficiency} = (MA / VR) \times 100 = (3.33 / 4) \times 100 = 83.3\%$

4. WAEC June 2012 (Paper 2, Q5b)

State two reasons why no machine is 100% efficient.

Answer:

- Energy is lost due to friction.
- Some energy is converted to heat or sound, not useful work.

5. NECO June 2020 (Paper 1, Q12)

The ratio of the load to the effort in a machine is known as:

- A. Mechanical advantage
- B. Velocity ratio
- C. Efficiency
- D. Power ratio

ASSIGNMENTS (WAEC/NECO STYLE)

1. A machine is used to lift a load of 900 N with an effort of 300 N. If the distance moved by the effort is 1.5 m and the load rises by 0.5 m, calculate:

- a. The mechanical advantage
- b. The velocity ratio
- c. The efficiency of the machine

2. Explain the following terms and give one example each:

- a. Simple machine
- b. Mechanical advantage
- c. Velocity ratio
- d. Efficiency

3. A lever is used to lift a heavy object. The effort arm is 2.5 m long while the load arm is 0.5m. calculate the velocity ratio and the efficiency of the lever if it has a mechanical advantage of 4.

4. Differentiate between a simple machine and a compound machine. Give one example of each used in everyday life.
5. A pulley system has a velocity ratio of 6. If the machine is 60% efficient, what load can be lifted by an effort of 100 N?



WEEK 6 - HEAT ENERGY: TEMPERATURE AND ITS MEASUREMENTS

MEANING OF HEAT AND TEMPERATURE

Heat Energy is a form of energy that flows from a region of higher temperature to a region of lower temperature due to a temperature difference. It is measured in **Joules (J)**.

Temperature is a measure of the degree of hotness or coldness of a body. It determines the direction of heat flow and is measured in **degrees Celsius (°C), Fahrenheit (°F), or Kelvin (K)**.

DIFFERENCE BETWEEN HEAT AND TEMPERATURE

Property	Heat	Temperature
Definition	Energy transferred due to a temperature difference	Degree of hotness or coldness
SI Unit	Joule (J)	Kelvin (K)
Type	Extensive property (depends on amount of matter)	Intensive property (independent of amount)
Instrument	Calorimeter	Thermometer

Thermometric Properties

Thermometers operate based on **physical properties** of materials that change with temperature. These properties are called **thermometric properties**.

Property	Instrument Example
Volume of liquid (e.g. mercury)	Mercury/alcohol thermometers
Electrical resistance	Resistance thermometer
Thermoelectric voltage	Thermocouple
Pressure of gas at constant volume	Gas thermometer
Colour change	Thermochromic strips

TYPES OF THERMOMETERS AND USES

1. Mercury-in-glass Thermometer: Mercury in glass thermometer can be calibrated by supporting a thermometer vertically ensuring the bulb is covered with melting ice shavings. Allow the thread to fall until it remains in a position. Then mark this point as your lower fixed point. Again suspend this thermometer in a steam of hypsometers, allow the thread to rise until it remains constant, mark this point as upper fixed point. Divide the interval to a number of equal parts.

- High precision
- Used in laboratories
- Range: -10°C to 110°C
- Not used in extreme cold

2. Alcohol-in-glass Thermometer:

- Used in very cold environments
- Alcohol expands more than mercury
- Colored for visibility
- Range: -80°C to $+78^{\circ}\text{C}$

3. Clinical Thermometer:

- Used to measure human body temperature
- Range: 35°C to 42°C
- Constriction prevents backflow of mercury

4. Digital Thermometers:

- Used in hospitals and homes
- Measures quickly
- Displays value electronically

5. Thermocouple Thermometer:

- Uses thermoelectric effect
- Suitable for high temperatures (up to 1800°C)
- Industrial applications

6. Bimetallic Strip Thermometer:

- Used in thermostats

- Consists of two metals bonded together that expand differently

7. Gas Thermometer:

- Uses pressure of gas at constant volume
- Very accurate

TEMPERATURE SCALES

1. Fundamental interval is the difference in temperature between the lower fixed point (temperature of melting ice at standard atmospheric pressure) and the upper fixed point (temperature of steam at standard atmospheric pressure).
2. Upper Fixed point is the temperature of steam from pure water boiling at standard atmospheric pressure of 760mm of mercury.
3. Lower Fixed point is the temperature of pure melting ice at standard atmospheric pressure of 760mm of mercury.

Celsius (Centigrade) Scale:

- Ice point = 0°C
- Steam point = 100°C
- Divided into 100 equal parts

Kelvin scale:

- Absolute scale of temperature
- Ice point = 273 K
- Steam point = 373 K

Fahrenheit Scale:

- Ice point = 32°F
- Steam point = 212°F

TEMPERATURE CONVERSION

1. Converting $^{\circ}\text{C}$ to Kelvin: $\text{K} = ^{\circ}\text{C} + 273$
2. Converting Kelvin to $^{\circ}\text{C}$: $^{\circ}\text{C} = \text{K} - 273$
3. Converting $^{\circ}\text{C}$ to Fahrenheit: $^{\circ}\text{F} = \frac{9^{\circ}\text{C}}{5} + 32$ or $^{\circ}\text{F} = 1.8^{\circ}\text{C} + 32$
4. Converting Fahrenheit: $^{\circ}\text{C} = \frac{5(^{\circ}\text{F}-32)}{9}$

SOLVING REAL-WORLD AND THEORETICAL PROBLEMS INVOLVING TEMPERATURE

Example 1:

An object is heated from 30°C to 57°C . The increase in its temperature on Kelvin scale is

Solution

Increase in temperature = $\theta_2 - \theta_1$;

θ_1 = initial temperature = 30°C ;

θ_2 = final temperature = 57°C

$$\Delta\theta = 57^{\circ}\text{C} - 30^{\circ}\text{C} = 27^{\circ}\text{C}$$

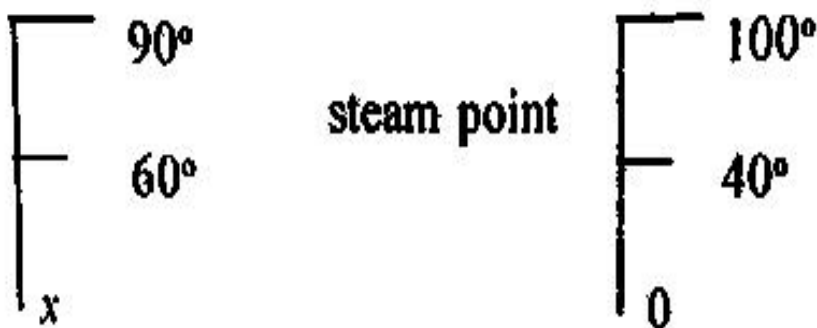
Applying the formula, $\text{K} = ^{\circ}\text{C} + 273$

$$\therefore \text{K} = 27 + 273 = 300 \text{ K}$$

Example 2:

The ice point of an ungraduated mercury-in-glass thermometer is X, while its steam point is 90° .

This thermometer reads 60° when the true temperature is 40°C . Calculate the value of X.



Solution

For any thermometer, the temperature scale is **linear**, so:

$$\frac{\text{Thermometer reading} - X}{\text{Steam point} - X} = \frac{\text{True temperature} - 0}{100 - 0}$$

Substitute values:

$$\frac{60 - X}{90 - X} = \frac{40}{100}$$

Cross-multiply:

$$100(60 - X) = 40(90 - X)$$

Expand:

$$6000 - 100X = 3600 - 40X$$

Rearrange:

$$6000 - 3600 = 100X - 40X$$

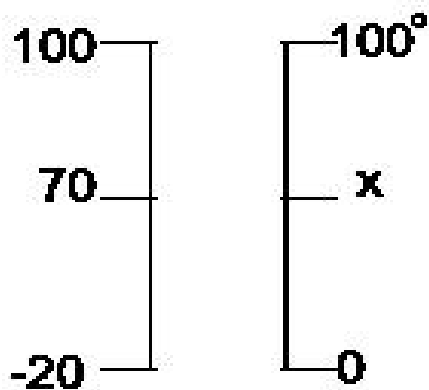
$$2400 = 60X$$

$$X = \frac{2400}{60}$$

$$X = 40^{\circ}\text{C}$$

Example 3:

The ice and steam point of a certain thermometer are -20° and 100° respectively. Calculate the Celsius temperature corresponding to 70° on the thermometers.



Solution

Using the linear scale relation:

$$\frac{T_c - 0}{100 - 0} = \frac{70 - (-20)}{100 - (-20)}$$

Simplify the right-hand side:

$$\frac{T_c}{100} = \frac{70 + 20}{120}$$

$$\frac{T_c}{100} = \frac{90}{120}$$

Cross-multiply:

$$T_c = \frac{90 \times 100}{120}$$

$$T_c = 75^\circ\text{C}$$

$$T_c = 75^\circ\text{C}$$

EVALUATION

1. What is the difference between temperature and heat?
2. State the SI unit of temperature.
3. Name and describe three types of thermometers.
4. Convert:
 - a) 25°C to K
 - b) 0°C to $^\circ\text{F}$
 - c) 273 K to $^\circ\text{C}$
5. What is a thermometric property?
6. Why is mercury preferred in thermometers?
7. List the fixed points used in thermometer calibration.

ASSIGNMENT

1. A thermometer reads 77°F . Convert this reading to:
 - a) $^\circ\text{C}$
 - b) Kelvin

2. State four advantages of alcohol over mercury in thermometers.
3. Draw and label a mercury-in-glass thermometer. Indicate the scale and bulb.
4. State five desirable properties of a liquid used in thermometers.
5. The wire of a platinum resistance thermometer has a resistance of 3.5Ω at 0°C and 10.5Ω at 100°C . Calculate the temperature of the wire when its resistance is 7.5Ω .



WEEK 8 & 9 - HEAT ENERGY MEASUREMENTS

DEFINITION OF HEAT ENERGY

- **Heat** is a form of energy that flows from a region of higher temperature to a region of lower temperature.
- It is transferred via conduction, convection, or radiation.
- **SI Unit:** Joule (J)
- Other unit: calorie (1 cal = 4.186 J)
- Heat is a scalar quantity.

SPECIFIC HEAT CAPACITY

This is the amount of heat required to raise the temperature of 1 kg of a substance by 1°C (or 1 K).

Formula:

$$Q = mc\theta$$

Where:

- Q = heat energy (J)
- m = mass (kg)
- c = specific heat capacity (J/kg·K)
- θ = temperature change

Examples of Specific Heat Capacities:

Substance	Specific Heat Capacity (J/kg·K)
Water	4200
Ice	2100
Aluminum	900
Copper	390
Iron	460

Example:

A box has a volume of 0.28m³ and is 70% filled with iron fillings at 25°C. Calculate the energy required to melt 10% of the iron fillings.

[Density of iron = 8.00 x 10³ Kg m⁻³, specific latent heat of fusion of iron = 1.38 x 10⁵ J Kg⁻¹, specific heat capacity of iron = 460 J Kg⁻¹ K⁻¹, melting point of iron = 1500°C]

Solution

$$\begin{aligned}\text{Energy} &= ml + mc\theta \\ &= (1568 \times 1.38 \times 10^5) + (1568 \times 460 + (1500 - 25)) \\ &= 216384000 + 1.06 \times 10^9 \\ &= 1.28 \times 10^9 \text{ J}\end{aligned}$$

LATENT HEAT

Latent heat is the amount of heat energy absorbed or released during a **change of state** without change in temperature.

Types:

1. **Latent Heat of Fusion (L_f):** Heat required to change 1 kg of a substance from solid to liquid at constant temperature.

$$Q = mL_f$$

2. **Latent Heat of Vaporization (L_v):** Heat required to change 1 kg of a substance from liquid to gas.

$$Q = mL_v$$

Units: J/kg

Table: Latent Heats of Common Substances

Substance	Latent Heat of Fusion (J/kg)	Latent Heat of Vaporization (J/kg)
Water	3.34×10^5	2.26×10^6
Ice	3.34×10^5	N/A
Alcohol	N/A	$\sim 8.5 \times 10^5$

CALORIMETRY: Calorimetric is the science of measuring the amount of heat exchanged in a physical or chemical process using a device called a **calorimeter**.

$$\text{Heat Gained} = \text{Heat Lost}$$

Example:

Calculate the heat required to raise 1.5 kg of water from 25°C to 100°C:

Solution

Using the formula, $Q = mc\theta$

$$= 1.5 \times 4200(100 - 25)^\circ\text{C}$$

$$= 6300 \times 75$$

$$= 472500\text{J}$$

METHODS OF MEASURING HEAT ENERGY

1. **Method of Mixtures:** Hot and cold substances mixed; final temperature measured.
2. **Electrical Method:** Heater supplies known power; time measured.
3. **Using Temperature Change:** Applying to known substance.

HEAT LOSS IN MIXING

Principle:

$$\text{Total Heat Lost} = \text{Total Heat Gained}$$

General Formula:

$$M_m C_m \Delta\theta_m = M_w C_w \Delta\theta_w + C_a \Delta\theta_a$$

Where:

- M_m = mass of the hotter substance
- C_m = specific heat capacity of the hotter substance
- $\Delta\theta_m$ = temperature change of the hotter substance
- M_w = mass of the colder substance
- C_w = specific heat capacity of the colder substance
- $\Delta\theta_w$ = temperature change of the colder substance
- C_a = heat absorbed by calorimeter (or surroundings)
- $\Delta\theta_a$ = temperature change of calorimeter

Example 1

A nursing mother prepared her baby's milk mixture at 85°C, in a feeding bottle. In order to cool it to 40°C, she immersed the bottle in an aluminium bowl of heat capacity 90 J K⁻¹ containing 500g of water at 26°C. If the mass of the mixture is 300g, calculate the specific heat capacity of the mixture.

[Neglect the heat losses and heat capacity of the bottle;

specific heat capacity of water = 4200 J kg⁻¹ K⁻¹]

Solution

Heat lost by mixture = Heat gained by water + Heat gained by aluminium

$$M_m C_m \Delta\theta_m = m_w C_w \Delta\theta_w + C_a \Delta\theta_a$$

$$0.3 \times C_m \times (85-40) = 0.5 \times 4200 \times (40-26) + 90 (40-26)$$

$$13.5 C_m = 29400 + 1260$$

$$C_m = 2271.1 \text{ J K}^{-1} \text{ K}^{-1}$$

Example 2:

A 40V electric heater is used to supply a current of 12A for 1400s to a body of mass 1.5kg at the melting point of the body. The body melts and its temperature rises through 60°C in an extra 72s. Determine the specific heat capacity of the body.





Step 3: Solve for C

$$C = \frac{34560}{90}$$

$$C = 384 \text{ J/kg} \cdot$$

Solution

Formula:

$$\text{Electrical Energy} = MC\theta$$

Where:

- $M = 1.5 \text{ kg}$
- $C = ?$
- $\theta = 60^\circ \text{C}$
- Power $P = IV = 40 \times 12 \text{ W}$
- Time for temperature rise = 72 s

Step 1: Calculate energy supplied

$$E = IVt$$

$$E = 40 \times 12 \times 72$$

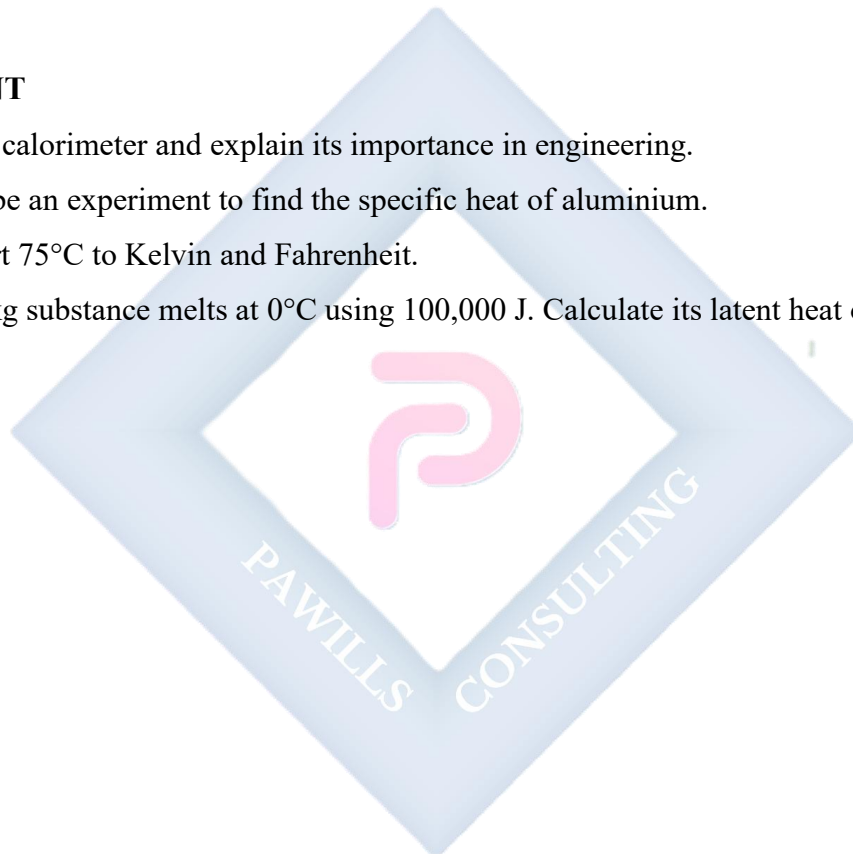
$$C = 384\text{J/kgK}$$

EVALUATION

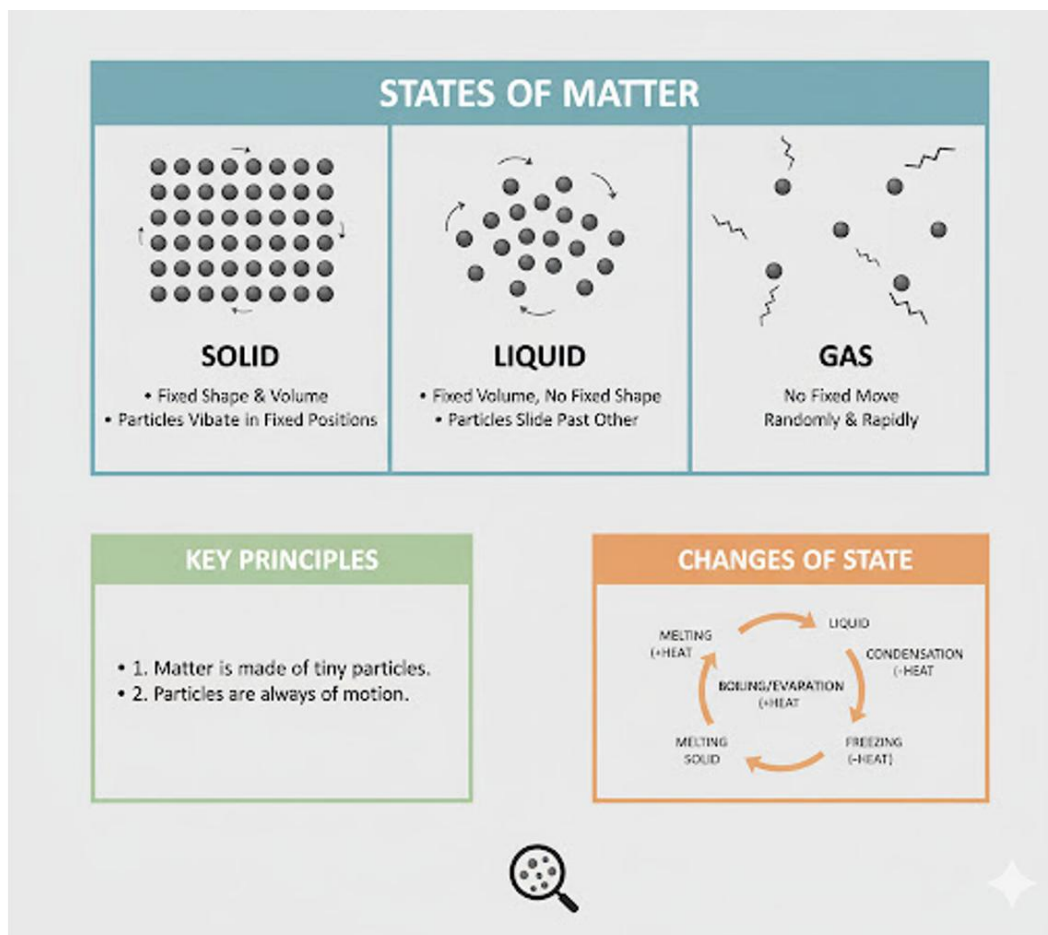
1. Define heat energy and list its units.
2. Differentiate between specific and latent heat.
3. A 2 kg metal requires 7200 J to rise from 20°C to 80°C. Find its specific heat capacity.
4. A 0.5 kg substance absorbs 167,000 J to vaporize. Find its latent heat.
5. List 4 real-life uses of calorimeter.

ASSIGNMENT

1. Define calorimeter and explain its importance in engineering.
2. Describe an experiment to find the specific heat of aluminium.
3. Convert 75°C to Kelvin and Fahrenheit.
4. A 0.3 kg substance melts at 0°C using 100,000 J. Calculate its latent heat of fusion.



WEEK 10 - PARTICULATE NATURE OF MATTER



DEFINITION AND NATURE OF MATTER

Matter is anything that has mass and occupies space. Matter exists in different states and is made up of tiny, indivisible particles known as atoms, molecules, or ions depending on the substance.

Particulate Nature of Matter:

The particulate nature refers to the concept that all matter is made up of small discrete units.

These particles are:

- Very small and cannot be seen with the naked eye.
- Continuously in motion.
- Held together by antiparticle forces.
- Separated by empty spaces (especially in gases).

STATES OF MATTER USING MOLECULAR THEORY

Property	Solid	Liquid	Gas
Particle Arrangement	Tightly packed in fixed positions	Loosely packed, can slide	Widely spaced, free-moving
Motion	Vibrate in place	Slide over one another	Move rapidly and randomly
Intermolecular Force	Very strong	Moderate	Weak
Shape	Definite	No definite shape	No definite shape
Volume	Definite	Definite	No definite volume

EXPLANATION OF CHANGES OF STATE:

- Heating gives particles energy → increases motion → change in state (e.g., melting = solid to liquid).

1. Diffusion:

The process by which particles spread from a region of high concentration to a region of low concentration.

Examples/Demonstration:

- Perfume spreading in a room.
- Potassium permanganate crystals placed in water spread without stirring.
- Ink dropped in water.

Explanation: Particles move and collide with each other, eventually spreading uniformly.

2. Brownian Motion:

The random and haphazard movement of tiny visible particles (like pollen grains or smoke) in a fluid due to collisions with smaller invisible molecules.

Demonstration/Experiment:

- Smoke cell observed under a microscope.

- Online simulation of Brownian motion.

Conclusion: This movement supports the theory that matter is made of constantly moving particles.

3. **Change of State:**

- Melting, boiling, evaporation, condensation, etc., all demonstrate molecular motion and energy.

KINETIC THEORY OF MATTER

The kinetic theory explains the behavior of matter in terms of the motion of its particles.

Assumptions of Kinetic Theory:

1. Matter is made up of small particles.
2. These particles are in constant, random motion.
3. There are spaces between the particles.
4. Collisions between particles and with the container walls are perfectly elastic (no energy loss).
5. The temperature of a substance is a measure of the average kinetic energy of its particles.
6. Intermolecular forces vary with state—strong in solids, weaker in liquids, and negligible in gases.

APPLICATIONS AND REAL-LIFE EXAMPLES

1. **Perfume spreads in air** – diffusion in gases.
2. **Sugar dissolving in water** – molecular interaction.
3. **Tyre expanding in heat** – gas molecules gain kinetic energy.
4. **Refrigerators** – based on latent heat and molecular motion.
5. **Cooking gas smell** – diffusion of gas particles.
6. **Boiling water** – molecules move faster and escape as steam.

EVALUATION

1. Define molecular theory of matter.
2. State and explain four assumptions of the theory.

3. Differentiate between atoms and molecules.
4. Describe the molecular behavior in solids, liquids, and gases.
5. Explain Brownian motion and what it proves.
6. Give two examples of diffusion in daily life.

ASSIGNMENT

1. List and explain five postulates of the molecular theory of matter.
2. Describe two experiments that prove the motion of molecules.
3. Draw a diagram to show the molecular arrangement in the three states of matter.
4. Explain how the kinetic energy of particles changes during heating and cooling.

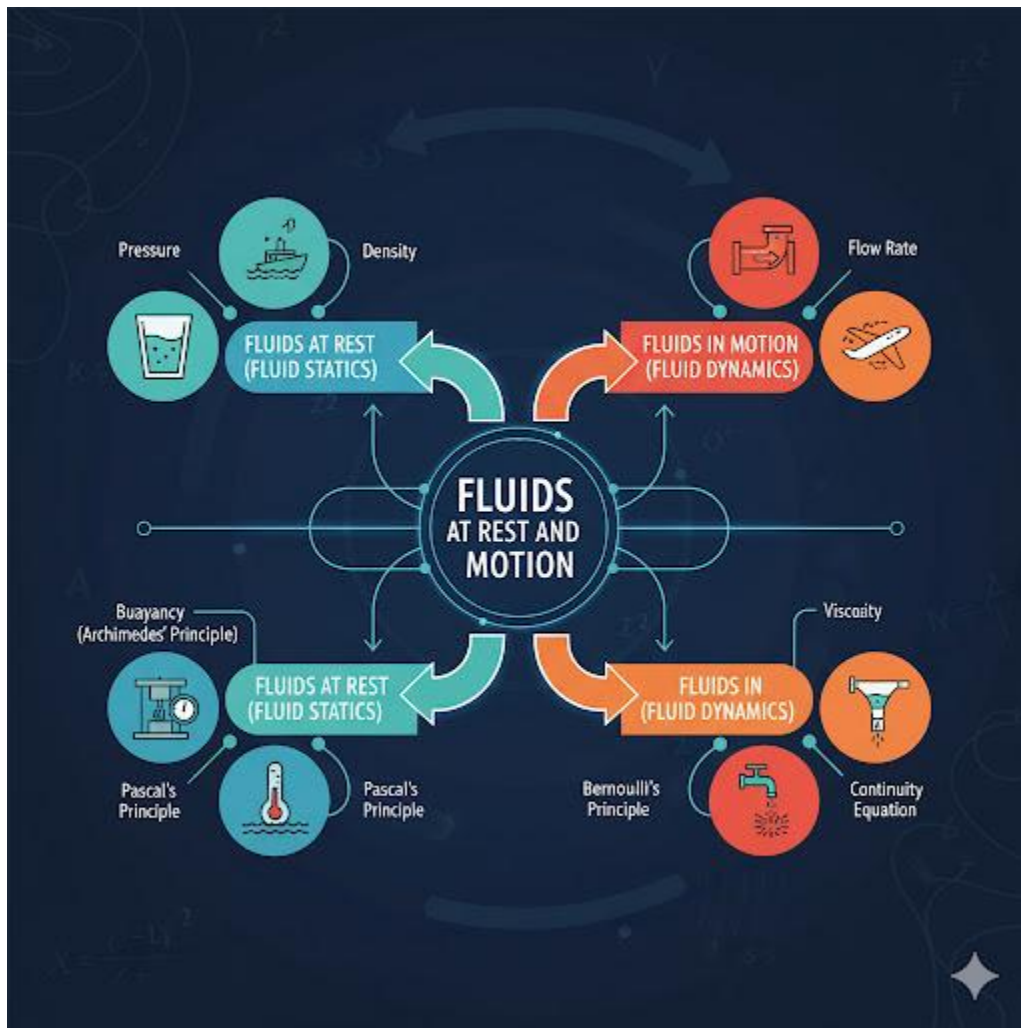
SS1 PHYSICS LESSON NOTES

THIRD TERM

1. Fluids at Rest and in Motion
2. Gas Laws
3. Simple Harmonic Motion
4. Projectile
5. Types of Waves
6. Properties of Waves



WEEK 1 - FLUIDS AT REST AND IN MOTION



DEFINITION AND TYPES OF FLUIDS

A fluid is any substance that can flow and take the shape of its container. This includes liquids and gases.

Key Properties of Fluids:

- They have no fixed shape.
- They exert pressure in all directions.
- They can be compressed (especially gases).
- They can flow (low internal resistance).

FLUIDS AT REST (FLUID STATICS)

Pressure in a Fluid at Rest

Formula:

$$P = \rho gh$$

Where:

- P = pressure (N/m^2 or Pascal)
- ρ = density of fluid (kg/m^3)
- g = acceleration due to gravity (9.8 m/s^2)
- h = depth from the fluid surface (m)

Characteristics:

- Pressure increases with depth.
- It acts equally in all directions at the same depth.
- Pressure is independent of shape or volume of the container.

FACTORS THAT AFFECT PRESSURE AT A POINT IN A LIQUID

1. Depth of the point **below the surface of the liquid**
2. Density of the liquid

PASCAL'S PRINCIPLE

Pressure applied to a confined fluid is transmitted equally and undiminished in all directions.

Mathematical Expression:

$$F_1/A_1 = F_2/A_2$$

Where:

- F_1 , F_2 = input and output forces
- A_1 , A_2 = areas of pistons

Applications:

- Hydraulic lifts
- Car brakes
- Jacks and presses
- Dentist chairs

ARCHIMEDES' PRINCIPLE

The principle states that a body wholly or partially submerged in a fluid experiences an upward force (upthrust) equal to the weight of the fluid displaced.

Formula for Upthrust:

$$F_u = \rho g V$$

Where:

- F_u = upthrust (N)
- ρ = density of fluid
- V = volume of fluid displaced

Apparent Weight:

Apparent weight = True weight – Upthrust

Applications:

- Design of ships and submarines
- Hydrometers used for measuring density
- Floating of ice on water

How to Determine the Relative Density Of A Stone Using Archimedes' Principle.

1. Weigh the stone in air (W_1).
2. Immerse the stone fully in water and weigh it again (W_2).
3. Upthrust = $W_1 - W_2$
4. Relative density = Weight in air / Upthrust = $W_1 / (W_1 - W_2)$

LAW OF FLOTATION

The Law of Floatation states that a body immersed in a fluid will float in the fluid when its weight is equal to the upthrust on it.

Conditions for Floating:

Weight of object = weight of fluid displaced

Explanation:

- An object will float if its density is less than the fluid's.
- It will sink if its density is greater.

Applications:

- Ship design
- Floating logs
- Life jackets
- Icebergs

FLUIDS IN MOTION (FLUID DYNAMICS)

1. Streamline and Turbulent Flow

- **Streamline Flow:** Flow in parallel layers without mixing (e.g., oil in a narrow pipe).
- **Turbulent Flow:** Irregular and chaotic flow (e.g., waterfall, fast rivers).

Laminar Flow Conditions:

- Low velocity
- Smooth surfaces
- Small diameter

2. Equation of Continuity

For incompressible, steady flow, the product of the cross-sectional area and velocity at any two points is constant.

$$A_1 V_1 = A_2 V_2$$

Implication:

- Narrower tubes → higher speed
- Wider tubes → lower speed

Application:

- Blood flow in arteries
- Fluid flow in irrigation and pipelines

3. Bernoulli's Principle

In a streamline flow of a fluid, the sum of pressure energy, kinetic energy, and potential energy per unit volume is constant.

$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{Constant}$$

Where:

- P = pressure
- ρ = fluid density
- v = velocity
- h = height

Applications:

- Airplane lift (wing design)
- Atomizers and sprayers
- Chimneys and carburetors
- Sailing

SURFACE TENSION: This is the force acting along the surface of a liquid, causing the liquid surface to behave like stretched elastic skin.

Effects of surface tension

These effects include:

1. Pond skater moving on water surface
2. Mercury droplets forming spheres
3. Needle floating in water

Methods by which the surface tension of a liquid can be reduced

- i. By adding alcohol
- ii. By adding alcohol
- iii. By adding camphor

Factors on which surface tension depends

The following factors affect surface tension:

- i. Nature of liquid
- ii. Temperature of liquid
- iii. Presence/Absence of Impurities

Work Examples

1. A uniform capillary tube, closed at one end contained dry air trapped by a thread of mercury $8.5 \times 10^{-2}\text{m}$ long. When the tube was held horizontally, the length of the air column was $5.0 \times 10^{-2}\text{m}$, when it was held vertically with the closed end downwards, the length was $4.5 \times 10^{-2}\text{m}$. Determine the value of the atmospheric pressure.
[$g = 10\text{ms}^{-2}$, density of mercury = $1.36 \times 10^4 \text{ kg m}^{-3}$]

Solution

Length of air thread of mercury = $8.5 \times 10^{-2}\text{m}$

When horizontal air column was $5.0 \times 10^{-2}\text{m}$

When vertical air column was $4.5 \times 10^{-2}\text{m}$

$g = 10\text{m/s}^2$, $p = 1.36 \times 10^4\text{kgm}^{-3}$

For horizontal,

$$e = 8.5 \times 10^{-2} - 5.0 \times 10^{-2} = 3.5 \times 10^{-2}\text{m}$$

$$P_1 = \rho gh$$

$$10 \times 1.36 \times 10^4 \times 3.5 \times 10^{-2}$$

$$4.76 \times 10^3\text{pascals}$$

For vertical,

$$L = 8.5 \times 10^{-2} - 4.5 \times 10^{-2} = 4.0 \times 10^{-2}\text{m}$$

$$P_2 = \rho gh$$

$$10 \times 1.36 \times 10^4 \times 4.5 \times 10^{-2}$$

$$5.44 \times 10^3\text{pascals}$$

$$\text{Atmospheric pressure} = 5.44 \times 10^3 + 4.76 \times 10^3$$

$$\Rightarrow 10.2 \times 10^3 \text{ pascals}$$

2. A body weighing 14N in air is partially immersed in water. If the mass of water displaced in the process is 200g, calculate the upthrust on the body. [$g = 10\text{ms}^{-2}$]

Solution

$$W = 14\text{N}$$

$$\text{mass of water displaced} = 200\text{g} = 0.2\text{kg}$$

$$\text{Upthrust} = \text{weight of the fluid displaced}$$

$$= \text{mass of the fluid displaced} \times \text{gravity}$$

$$= 0.2\text{kg} \times 10\text{m/s}^2$$

$$= 2\text{N}$$

3. A hydraulic press has a small piston of area 0.01 m^2 and a large piston of area 0.5 m^2 . If a force of 200 N is applied to the small piston, calculate the force exerted by the large piston.

Solution

Using Pascal's Principle:

$$P = F_1 / A_1 = F_2 / A_2$$

$$\text{So, } F_2 = (F_1 \times A_2) / A_1 = (200 \times 0.5) / 0.01 = \mathbf{10,000\text{ N}}$$

EVALUATION

Class Work (Oral & Written):

1. Define a fluid and give 3 examples.

2. NECO June 2016 (Paper 2, Q6)

State two factors that affect pressure at a point in a liquid.

3. WAEC June 2008 (Paper 1, Q8)

The pressure at a depth h in a liquid of density ρ is given by which of the following expressions?

A. $P = \rho gh$

B. $P = \rho g/h$

C. $P = h/\rho g$

D. $P = \rho/hg$

4. NECO June 2020 (Paper 2, Q3)

Describe how you would determine the relative density of a stone using Archimedes' principle.

5. Calculate the pressure at a depth of 5 m in water ($\rho=1000\text{kg/m}^3$).

6. State Bernoulli's principle and explain its use in airplane wings.

7. WAEC June 2012 (Paper 1, Q15)

Which of the following statements is correct about pressure in a liquid at rest?

- A. It decreases with depth
- B. It acts only downward
- C. It is the same in all directions at a point
- D. It acts only at the surface

ASSIGNMENT

1. A metal cube of volume 0.002 m^3 is immersed in a liquid of density 1200 kg/m^3 .

Calculate:

- a. The upthrust.
- b. The apparent weight if the weight in air is 25 N.

2. 1. WAEC June 2016 (Paper 2, Q4a)

State Archimedes' principle and list two applications.

3. 2. NECO June 2018 (Paper 2, Q5b)

A block of wood weighs 4.0 N in air and 3.0 N when fully immersed in water. Calculate:

- i. The upthrust on the block
- ii. The relative density of the block

4. A pipe has cross-sectional areas of 0.04 m^2 and 0.01 m^2 at two ends. If water enters at 3 m/s, calculate the exit velocity.

5. A liquid of volume 2.00 m^3 and density $1.00 \times 10^3\text{ kg m}^{-3}$ is mixed with 3.00 m^3 of another liquid of density $8.00 \times 10^3\text{ kg m}^{-3}$. Calculate the density of the mixture.

[Assume there is no chemical reaction]

WEEK 2 & 3 - GAS LAWS

DEFINITION OF A GAS

A gas is a state of matter that has **no fixed shape or volume**. It takes the shape and volume of its container. Gas particles are widely spaced and move in random, rapid motion.

ASSUMPTIONS OF KINETIC THEORY OF GASES

1. Gases are made of large numbers of particles (atoms or molecules) in random motion.
2. The volume of the particles is negligible compared to the volume of their container.
3. There are no forces of attraction or repulsion between the particles.
4. Collisions between gas particles and container walls are perfectly elastic.
5. The average kinetic energy of gas molecules is proportional to absolute temperature.

$$KE_{avg} = \frac{3}{2} kT$$

Where:

- $K = 1.38 \times 10^{-23} \text{ J/K}$ (Boltzmann constant)
- T = Temperature in Kelvin

BOYLE'S LAW (PRESSURE-VOLUME LAW)

This law states at constant temperature, the volume of a fixed mass of gas is **inversely proportional** to its pressure.

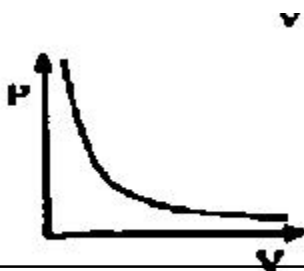
Mathematically;

$$P \propto \frac{1}{V} \text{ or } PV = k$$

Derived Equation (for comparison):

$$P_1 V_1 = P_2 V_2$$

Graphical Representation:

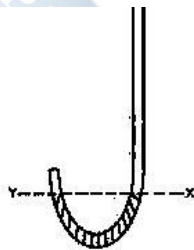
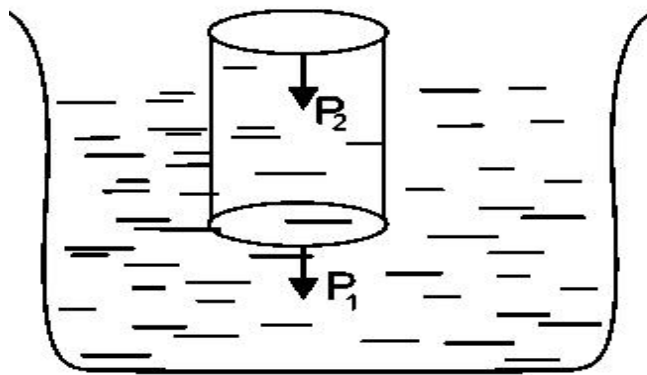


Pressure vs Volume → Hyperbolic curve

Pressure vs $1/\text{Volume}$ → Straight line

Experimental Demonstration:

Use a syringe sealed at the nozzle. Push the plunger while observing the pressure change.



Work Example:

In the J-tube above, Y and X are on the same horizontal level and 30cm^3 of air is trapped above Y when the atmospheric pressure is 75cm Hg , calculate the volume of air trapped above Y when 15cm Hg is now poured into the limb above X.

Solution

Given: the process is isothermal (temperature constant), so

$$P_1 V_1 = P_2 V_2.$$

Let $P_1 = 75$, $V_1 = 30$, and the final pressure $P_2 = 75 + 15 = 90$.

(Ensure pressure and volume use consistent units — here we keep the same units throughout so they cancel.)

Working

$$P_1 V_1 = P_2 V_2$$

$$75 \times 30 = 90 \times V_2$$

$$V_2 = \frac{75 \times 30}{90}$$

Simplify:

$$\frac{75}{90} = \frac{5}{6}, \quad \frac{5}{6} \times 30 = 25$$

Answer

$$V_2 = 25 \text{ (volume units)}$$

Note: If V_1 was given in cm^3 , then $V_2 = 25 \text{ cm}^3$. The final unit should match whatever unit V_1 was originally expressed in.

CHARLES'S LAW (VOLUME-TEMPERATURE LAW)

The law states that at constant pressure, the volume of a fixed mass of gas is **directly proportional** to its absolute temperature.

Mathematically:

$$V \propto T \Rightarrow \frac{V}{T} = \text{constant}$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Note: Temperatures unit must be in **Kelvin**.

Work Example

A gas has a volume of 100 cm^3 at 27°C .

It is heated until its volume becomes 120 cm^3 .

Find the new temperature T .

Solution

Using Charles' Law:

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Where:

- $V_1 = 100 \text{ cm}^3$
- $V_2 = 120 \text{ cm}^3$
- $T_1 = 27^\circ\text{C} = 27 + 273 = 300 \text{ K}$
- $T_2 = ?$

Substitute the values

$$\frac{100}{300} = \frac{120}{T_2}$$

Cross-multiply:

$$100T_2 = 120 \times 300$$

$$T_2 = \frac{120 \times 300}{100}$$

$$T_2 = 360 \text{ K}$$

$$T = 87^\circ\text{C}$$

Convert back to Celsius

$$T = 360 - 273 = 87^\circ\text{C}$$

PRESSURE LAW (GAY-LUSSAC'S LAW)

The law states that at constant volume, the pressure of a fixed mass of gas is directly proportional to its absolute temperature.

Mathematically:

$$P \propto T \Rightarrow \frac{P}{T} = \text{constant}$$

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

Demonstration:

Seal a can with some air inside and heat it. Observe pressure increase or bursting due to increased pressure.

Work Example:

A fixed mass of gas at a temperature of 27°C is heated at constant volume until the pressure is double. The final temperature of the gas is?

Solution

$$T_1 = 27 + 273 = 300\text{K}$$

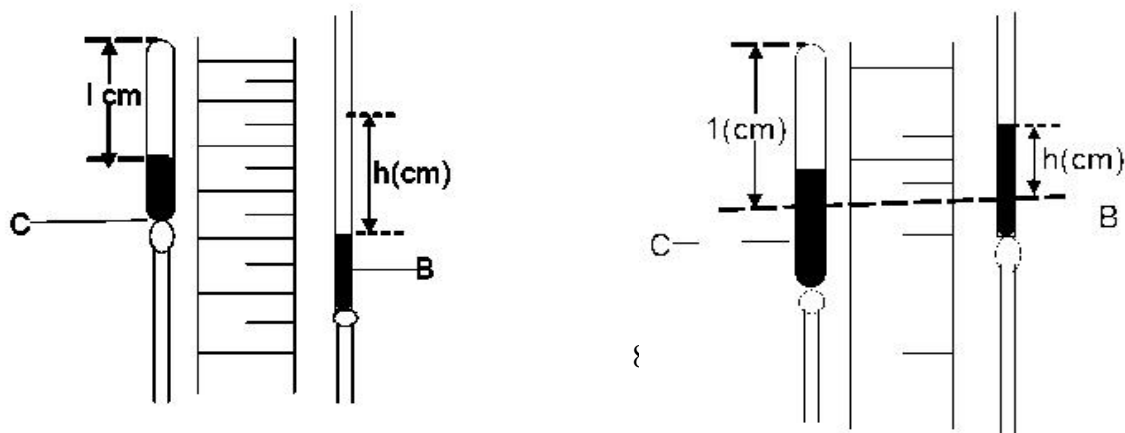
$$\frac{T_2}{T_1} = \frac{P_2}{P_1} = 2$$

$$T_2 = 2T_1 = 2 \times 300 = 600\text{K} = 327^\circ\text{C}$$

GENERAL GAS LAW

Combining Boyle's, Charles's and Pressure Laws:

This is the relationship between the volume and pressure of a given mass of gas at constant temperature.



Atmospheric pressure = $H \text{ cmHg}$

$$P = (H - h) \text{ cmHg}$$

Atmospheric pressure = $H \text{ cmHg}$

$$P = (H + h) \text{ cmHg}$$

Some dry air is enclosed in a glass tube of uniform cross-section attached by rubber tubing to a mercury reservoir.

By raising or lowering the open tube (B), the length of the air column in the closed tube (C) can be varied. The length read on the metre scale is measured as the volume of the gas (since tube is of uniform cross-section). The difference (h) in the levels of mercury at C and B is read for each value in length of gas.

The atmospheric pressure (H) is read from a fortin barometer.

When B is lower than C, pressure is,

$$P = (H - h) \text{ cmHg}$$

When C is higher than B, pressure is

$$P = (H + h) \text{ cmHg}$$

various values of l and h is obtained by lowering and raising tube B. A graph of P is plotted against V(Lcm). It gives a result that shows it varies inversely.

$$PVT = \text{constant}$$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

Used when pressure, volume, and temperature all change simultaneously.

Work Example:

The volume and pressure of a given mass at 27°C are 76cm^3 and 80cm of mercury respectively. calculate its volume at s.t.p.

Solution

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$T_1 = 27^\circ\text{C} = 27 + 273 = 300\text{K}$$

$$P_1 = 80\text{cmHg} = 800\text{mmHg}$$

$$V = 76\text{cm}^3$$

$$T_2 = 273\text{K}$$

$$P_2 = 760\text{mmHg}$$

$$V_2 = ?$$

Making V_2 the subject of formula

$$V_2 = \frac{P_1 V_1 T_2}{P_2 T_1}$$

$$V_2 = 800 \times \frac{76}{760} = 72.8\text{cm}^3$$

Ideal Gas Equation

$$PV = nRT$$

Where:

- P = pressure (Pa)
- V = volume (m^3)
- n = number of moles
- R = universal gas constant = $8.314 \text{ J/mol}\cdot\text{K}$
- T = temperature (K)

Used for:

- Gases involving moles
- Advanced calculations and molar mass estimations

MOLAR VOLUME AND AVOGADRO'S LAW

Avogadro's Law:

The law states that equal volumes of all gases, at the same temperature and pressure, contain equal numbers of molecules.

Molar Volume:

At STP (Standard Temperature = 273 K, Standard Pressure = 1 atm),

1mol of gas = 22.4dm³ or 22.4L

Work Example:

The pressure of one mole of an ideal gas of volume 10⁻²m³ at a temperature of 27⁰C is

[Molar gas constant = 8.3 Jmol⁻¹K⁻¹]

Solution

Parameters given:

Pressure, P =?

Volume, V = 10⁻²m³

Temperature, T = 27⁰C

Molar gas constant = 8.3Jmol⁻¹k⁻¹

Recall: $Pv = nRT$

$$\therefore P = \frac{nRT}{V}$$

$$P = 1.0 \times 8.3 \times \frac{300}{10^{-2}}$$

$$P = 300 \times 102 \times 8.3$$

$$= 8.3 \times 3 \times 10^4$$

$$= 24.9 \times 10^4$$

$$= 2.49 \times 10^5$$

$$= 2.49 \times 10^5 \text{ Nm}^{-2}$$

REAL VS IDEAL GASES

Ideal Gas Assumptions:

- No intermolecular forces

- Collisions are perfectly elastic
- Volume of particles negligible

Real Gases:

- Deviate at **high pressure** and **low temperature**
- Explained using **Van der Waals equation**

10. Applications of Gas Laws

- Automobile airbags (Boyle's law)
- Hot air balloons (Charles's law)
- Pressure cookers (Pressure law)
- Weather balloons
- Breathing and lung function (Boyle's law)

Evaluation

1. State three assumptions of kinetic theory of gases.
2. Derive the general gas law from the basic gas laws.
3. A gas of volume 3 L at 300 K and 2 atm is heated to 600 K and compressed to 4 atm. Find its final volume.
4. Explain why gases deviate from ideal behavior at high pressure.
5. Define molar volume and state Avogadro's Law.

Assignment

1. Research and summarize the **Van der Waals Equation**.
2. A balloon contains 5.6 g of helium. How many moles is this? (He = 4 g/mol)
3. A gas at STP has a volume of 44.8 L. How many moles are present?
4. If 0.5 mol of gas exerts a pressure of 2 atm at 300 K, calculate the volume. Use $R=0.0821 \text{ atm}$.
5. The volume of a given mass of an ideal gas at 327K and $9.52 \times 10^4 \text{ Pa}$ is 40 cm^3 . Calculate the volume of the gas at 273K and $1.034 \times 10^5 \text{ Pa}$.

WEEK 4 & 6 - SIMPLE HARMONIC MOTION

DEFINITION OF SIMPLE HARMONIC MOTION

Simple Harmonic Motion is a special type of periodic motion in which the acceleration of a particle is directly proportional to its displacement from a fixed point (equilibrium position) and is always directed towards that point.

$$a \propto -x \Rightarrow a = -\omega^2 x$$

Where:

- a = acceleration
- x = displacement from the mean position
- ω = angular frequency
- The negative sign indicates that the acceleration is directed toward the equilibrium (restoring force).

The magnitude of the force required to make an object of mass M move with a speed V in a circular path of radius R is known as the centripetal force and it is given by the expression:

$$F = \frac{mV^2}{r}$$

CONDITIONS FOR SHM

A system exhibits SHM if:

- There is a **restoring force** that acts toward the equilibrium position.
- The magnitude of this force is **proportional** to the displacement.
- The motion is **repetitive** and oscillatory.

Examples of SHM Systems

- A simple pendulum (for small angles)
- A mass-spring system (horizontal or vertical)
- Vibrating tuning fork
- Oscillating swing
- Electrical oscillations in LC circuits (at advanced level)

Factors That Affect SHM

- The length of the pendulum(l) and

- The strength of the gravitational field(g)

MATHEMATICAL EQUATIONS OF SHM

Quantity	Symbol	Equation
Displacement	X	$x = A\sin(\omega t)$ or $x = A\cos(\omega t)$
Velocity	V	$v = \omega\sqrt{A^2 - x^2}$ or $v = \frac{dy}{dx}$
Acceleration	A	$a = -\omega^2 x$ or $a = -\omega^2 A$
Angular velocity	Ω	$\omega = 2\pi f$ or $\omega = \sqrt{\frac{m}{k}}$
Period	T	$T = \frac{2\pi}{\omega}$ or $T = \frac{1}{f}$
Frequency	F	$F = \frac{1}{T}$

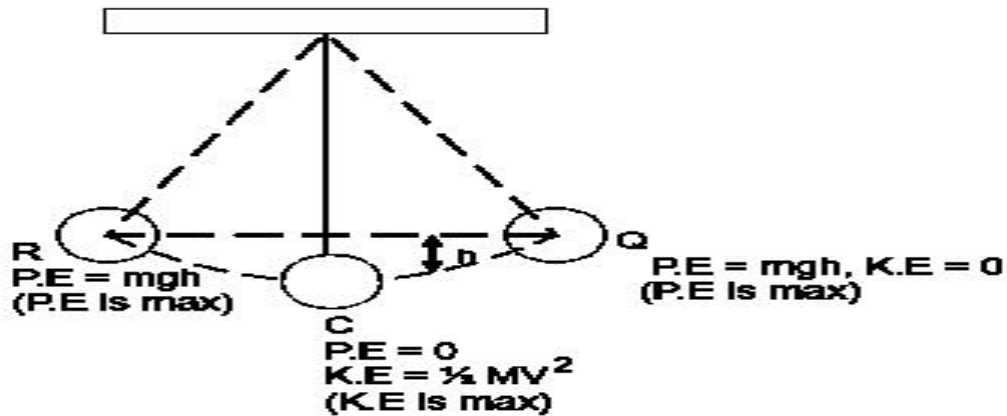
Where:

- a = amplitude (maximum displacement)
- t = time (s)
- ω = angular velocity (rads^{-1})
- k = spring constant (N)
- m = mass (kg)

Key Terms In SHM

Term	Definition
Amplitude (A)	Maximum displacement from the mean position
Period (T)	Time taken for one complete oscillation
Frequency (f)	Number of oscillations per second
Phase	Describes the position of a particle at a given instant
Angular velocity (ω)	$\omega = 2\pi f$ or $\omega = \frac{2\pi}{T}$

ENERGY IN SHM



The energy of a simple harmonic motion is derived when the motion of a loaded spiral spring is considered. When the motion of an object is S.H.M, there is always a restoring force tending to return the object to its central equilibrium position.

Kinetic Energy (K.E.):

$$K.E = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2 A^2$$

Potential Energy (P.E.):

$$P.E = mgh, \quad P.E = K.E$$

$$mgh = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2 A^2$$

SIMPLE PENDULUM

The motion of a simple pendulum is a typical example of a simple harmonic motion. All equations given above can be applied. The period of a simple pendulum is given as

$$T = 2\pi\sqrt{\frac{l}{g}}$$

Where;

T = periods of oscillation (s)

l = length of pendulum (m)

g = acceleration due to gravity (m/s^2)

Work Example

The period of a simple pendulum of length 80.0cm was found to have doubled when the length of the pendulum was increased by X . Calculate X .

Solution

Given: (1) The length, l , of the pendulum = 80.00cm

(2) Length, $l_2 = L_1 + x$

(3) Initial period = T_1

(4) Final period = $2T_1$ (doubled)

Recall: $T \propto \sqrt{L}$

$$\begin{array}{ccc} T_1 & T_2 \\ \hline \therefore \frac{\quad}{\sqrt{L_1}} & \frac{\quad}{\sqrt{L_2}} \\ = & \end{array}$$

$$\begin{array}{ccc} T_1 & 2T_1 \\ \hline \therefore \frac{\quad}{\sqrt{80}} & = \frac{\quad}{\sqrt{L_2}} \\ \quad & \quad \end{array}$$

$$\therefore \frac{T_1}{\sqrt{L_1}} = \frac{T_2}{\sqrt{L_2}}$$

$$\frac{T_1}{\sqrt{80}} = \frac{2T_1}{\sqrt{L_2}}$$

$$\therefore \frac{1}{\sqrt{80}} = \frac{2}{\sqrt{L_2}}$$

$$\sqrt{\frac{L_2}{80}} = \frac{2}{1}$$

$$\therefore \frac{x}{80} = \frac{4}{1}$$

$$x = 320 \text{ cm}$$

$$\text{Since } L_2 = L_1 + x$$

$$\therefore 320 = 80 + x$$

$$x = 240 \text{ cm.}$$

SAMPLE PROBLEMS

Example 1:

The surface of an ear drum (assumed circular) has a radius 2.1mm. It resonates with an amplitude of $0.8 \times 10^{-7} \text{ m}$ as a result of impulses received from an external body vibrating at 2400Hz. If the resulting pressure change on the ear drum is $3.6 \times 10^{-5} \text{ Nm}^{-2}$, calculate the acceleration. [$\pi = 3.142$]

solution

Acceleration

$$a = \omega^2 r = 2 \pi f r$$

$$= 2 \times 3.14 \times 2400 \times 2.1 \times 10^{-7}$$

$$= 18.2 \text{ ms}^{-2}$$

Example 2

An object of mass 0.40 kg attached to the end of a string is whirled round in a horizontal circle of radius 2.0 m with a constant speed of 8 ms^{-1} . Calculate the angular velocity of the object.

Solution

$$v = \omega r; v = 8 \text{ ms}^{-1}; r = 2.0 \text{ m}; \omega = ?$$

$$v = 8$$

$$\Omega = \frac{v}{r} = \frac{8}{2} = 4.0 \text{ rads}^{-1}$$

Example 3:

A sonometer wire vibrates in simple harmonic motion with a maximum amplitude of 1.0cm. calculate the frequency of vibration of the wire, given that the magnitude of the maximum

acceleration of the wire is 988ms^{-2} .

$[\pi = 22/7]$

Solution

$$a = \omega^2 r; \omega = 2\pi f$$

$$a = (2\pi f)^2 r$$

$$988 = (2 \times 22/7 \times f)^2 \times 1.0/100$$

$$f = 50.0\text{Hz}$$

Applications of SHM

- Designing clocks (pendulum and quartz)
- Vehicle suspension systems
- Shock absorbers
- Seismographs
- Musical instruments (strings, tuning forks)

Evaluation

1. Define SHM and state its conditions.
2. List and explain any four terms used in SHM.
3. A particle performs SHM with amplitude 10 cm and frequency 2 Hz. Write the equation of motion assuming it passes through the mean position at $t=0$.
4. Calculate the period of a simple pendulum of length 1.5 m. (Take $g=9.8\text{m/s}^2$)
5. Draw the displacement-time graph of SHM.
6. State two factors that affect the period of oscillation of a simple pendulum.

Assignment

1. Derive the expression for the total mechanical energy of a body in SHM.
2. A body of mass 0.5 kg oscillates with a frequency of 4 Hz and amplitude of 6 cm. Find:
 - a) Angular frequency

- b) Maximum speed
- c) Maximum acceleration
- 3. Research: Why does a pendulum clock go faster in valleys and slower on mountains?
- 4. During a training session, two footballers pass a ball repeatedly between themselves.
Give two reasons why the to and fro motion of the ball is not simple harmonic



WEEK 8 - PROJECTILE

DEFINITION OF PROJECTILE

A projectile is any object that once projected or dropped continues in motion by its own inertia and is influenced only by the downward force of gravity. or

It is an object that is propelled forward and then moves under the influence of gravity and air resistance. It's commonly associated with objects launched in warfare or sports, like bullets or baseballs. For an object to be considered a projectile, an initial force must be applied to it, after which it moves freely, with gravity and air resistance being the primary forces acting upon it.

Examples of projectiles are:

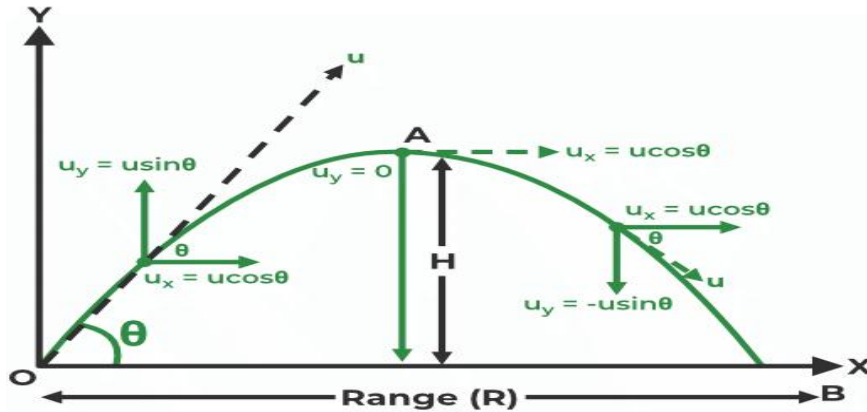
- (i) A stone thrown vertically upwards.
- (ii) A ball rebounding from a wall.
- (iii) An athlete doing high jump.
- (iv) A stone release from a catapult.
- (v) A bullet fired from a gun.

PROJECTILE MOTION

Projectile motion is a form of motion where an object moves in a parabolic path. The path followed by the object is called its trajectory. Projectile motion refers to the movement of an object thrown or projected into the air and influenced by gravity.

Projectile motion can be considered under three headings as follows:

- (a) Motion of an object thrown vertically into the air.
- (b) Motion of an object projected horizontally from a certain height.
- (c) Motion of an object projected into the air at an angle of inclination to the horizontal/vertical.



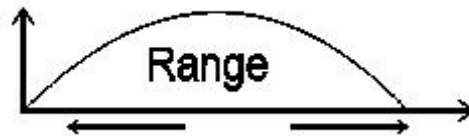
In a Projectile Motion, there are two simultaneous independent rectilinear motions:

1. Along the x-axis: uniform velocity, responsible for the Horizontal (forward) motion of the particle.
2. Along the y-axis: uniform acceleration, responsible for the Vertical (downwards) motion of the particle.

TERMS USED IN PROJECTILE MOTION

Range(R): This is defined as the horizontal distance from the point of projection to the point where the projectile hits the projection plane again.

$$R = \frac{U^2 \sin 2\theta}{g}$$



Maximum height (H): This is defined as the highest vertical distance reached as measured from the horizontal projection plane.

$$H = \frac{(u \sin \theta)^2}{2g}$$

Time of flight (t): The time taken by the projectile to complete its trajectory is called time of flight.

$$T = \frac{2u \sin \theta}{g}$$

Quantity	Formula
Time of flight	$T = \frac{2u \sin \theta}{g}$
Maximum height	$H = \frac{(u \sin \theta)^2}{2g}$
Range	$R = \frac{U^2 \sin 2\theta}{g}$
Vertical displacement	$Y = U \sin \theta \cdot t - \frac{1}{2}gt^2$
Horizontal displacement	$X = U \cos \theta \cdot t$

TRAJECTORY OF A PROJECTILE

The trajectory of a projectile is the curved path it follows through the air after being launched. Under ideal conditions (no air resistance), this path is parabolic. However, the actual shape and distance of the trajectory are influenced by several factors.

Below are the key factors that affect a projectile's trajectory:

Factor	Effect on Trajectory
Initial velocity	Increases range and height
Angle of projection	Affects shape, range, and height; 45° gives maximum range
Gravity (g)	Greater gravity shortens range and height

Height of launch	Higher launch point increases time of flight and range
Air resistance	Decreases height and range; alters path shape
Mass (in real life)	Heavier projectiles resist drag better
Shape & spin	Changes trajectory due to lift or drag (e.g., curved football shot)

WORK EXAMPLE RANGE, MAXIMUM HEIGHT, AND TIME OF FLIGHT

Example 1:

The ratio of maximum range to maximum height of a projectile is 4. Determine the angle of projection.

Solution

$$\text{Maximum range, } R = \frac{U^2 \sin 2\theta}{g}$$

$$\text{Maximum height, } H = \frac{U^2 \sin^2 \theta}{2g}$$

$$\text{Since } R : H = 4$$

$$\Rightarrow \frac{U^2 \sin 2\theta}{g} : \frac{U^2 \sin^2 \theta}{2g} = 4$$

$$\Rightarrow \frac{U^2 \sin 2\theta}{g} : \frac{U^2 \sin^2 \theta}{2g} = 4$$

$$\Rightarrow \frac{2 \sin 2\theta}{\sin^2 \theta} = 4$$

$$\Rightarrow \frac{U^2 \sin 2\theta}{g} : \frac{2g}{U^2 \sin 2\theta} = 4$$

$$\text{since } \sin 2\theta = 2 \sin \theta \cos \theta$$

$$\Rightarrow 2 \times \frac{2\sin\theta\cos\theta}{\sin^2\theta} = 4$$

$$\Rightarrow \frac{\sin\theta\cos\theta}{\sin^2\theta} = 1$$

$$\Rightarrow \frac{4\sin\theta\cos\theta}{\sin^2\theta} = 4$$

$$\Rightarrow \frac{\cos\theta}{\sin\theta} = 1$$

$$\Rightarrow \frac{\sin\theta}{\cos\theta} = 1$$

$$\Rightarrow \tan\theta = 1$$

$$\Rightarrow \theta = \tan^{-1}(1)$$

$$\therefore \theta = 45^\circ$$

Example 2:

An object is projected with a velocity of 80ms^{-1} at an angle of 30° to the horizontal. The maximum height reached is.

Solution

The formula for maximum height is:

$$H = \frac{U^2 \sin^2 \theta}{2g}$$

Given:

- $U = 80 \text{ m/s}$
- $\theta = 30^\circ$
- $g = 10 \text{ m/s}^2$

Step 1: Compute $\sin 30^\circ$

$$\sin 30^\circ = \frac{1}{2}$$

$$\sin^2 30^\circ = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

Step 2: Substitute values

$$H = \frac{80^2 \times \frac{1}{4}}{2 \times 10}$$

$$H = \frac{6400 \times \frac{1}{4}}{20}$$

$$H = \frac{1600}{20}$$

$$H = 80 \text{ m}$$

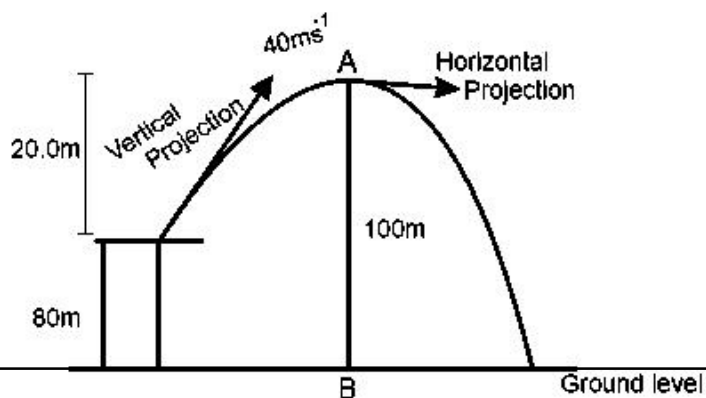
Example 3:

An object is projected from a height of 80m above the ground with a velocity of 40ms⁻¹ at an angle of 30° to the horizontal. The time of flight is

[$g = 10 \text{ ms}^{-2}$]

Solution

Diagrammatically,



$$\begin{aligned}
 T &= \frac{2u \sin \theta}{g} \\
 &= 2 \times 40 \times \frac{\sin 30^\circ}{10} \\
 &= \frac{40}{10} \\
 &= 4\text{s}
 \end{aligned}$$



EVALUATIONS

Objective Questions:

1. A projectile has zero vertical velocity at what point?
 - a) Initial point
 - b) Midpoint
 - c) Maximum height
 - d) Final point
2. The range of a projectile is maximum at an angle of:
 - a) 90°
 - b) 60°
 - c) 30°
 - d) 45°

Theory Questions:

1. Define projectile motion and give four examples.
2. Derive the formula for horizontal range.
3. A ball is thrown at 20 m/s at 60° . Calculate time of flight, height, and range.
4. A projectile has a time of flight of 6 s and reaches a maximum height of 45 m. Find its initial speed and launch angle.

ASSIGNMENT

1. A stone is projected from the ground at an angle of 60° and lands 100 m away. Find:
 - a) Time of flight
 - b) Initial speed
2. Draw and label the trajectory of a projectile. Indicate:
 - (i) Launch angle
 - (ii) Maximum height
 - (iii) Range
 - (iv) Time of flight
 - (v) Velocity components
3. A football is kicked at 15 m/s at an angle of 40° . Find the time to reach max height and the total range.

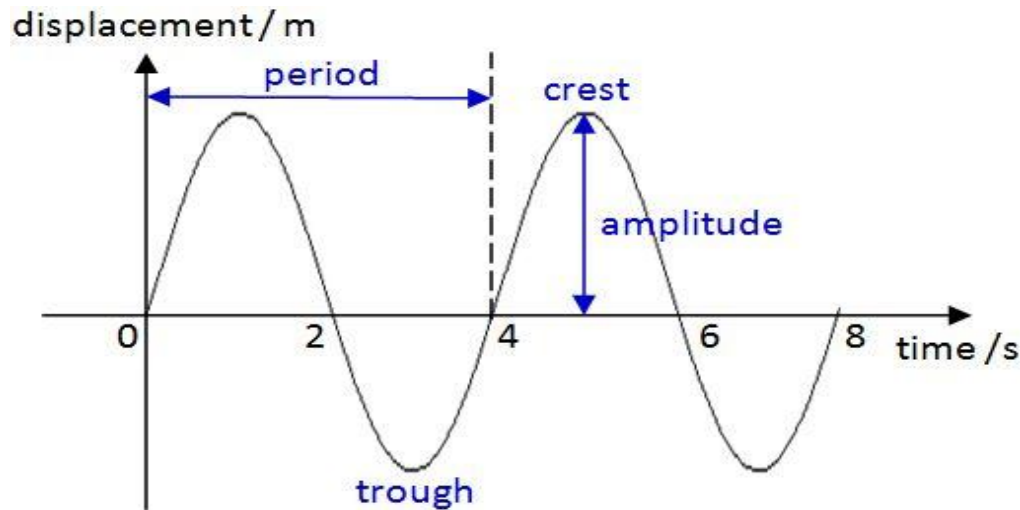


WEEK 9 - TYPES OF WAVES

DEFINITION OF WAVE

A wave is a **disturbance** that travels through a **medium or space**, carrying **energy** without the transport of matter.

- Waves originate from **vibrations**.
- Waves are everywhere: light, sound, water waves, etc.



DIFFERENT PARTS OF A WAVE:

Here are the most important parts of a wave.

- **Rest position** – This is the position of particles when there is no energy or wave passing through them; these particles are undisturbed. On a wave diagram, this is the line that is drawn through the centre of the wave.
- **Displacement** – This is the distance that a particle has moved from its rest position due to its disturbance.
- **Amplitude** – The measurement of displacement, we call the amplitude the maximum disturbance of a particular point in the medium. This is the distance between the rest position and the highest peak or trough.
- **Peaks and troughs** – The points of greatest disturbance or maximum displacement, above and below the rest position.
- **Oscillation** – The repetition of a wave – i.e. the space from peak to peak.

- **Wavelength** – The actual distance covered by one oscillation – usually measured from peak to peak.
- **Frequency or wave speed** – The number of times a wave oscillates in a second.
- **Period:** The time taken for one complete wave to pass a given point.
- **Unit:** Seconds (s)

Formula: $T = \frac{1}{f}$

Wave Speed (v): The speed at which the wave travels through a medium.

- **Formula:** $v = f \times \lambda$
- **Unit:** metres per second (m/s)

Where:

- v = wave speed (m/s)
- f = frequency (Hz)
- λ = wavelength (m)

Phase: The relative position of a point within a wave cycle.

- **Same phase:** Points that have the same displacement and direction of motion.
- **Out of phase:** Points that are not aligned in displacement and motion.

CLASSIFICATION OF WAVES

(i) Based on Requirement of Medium:

Type of Wave	Requires Medium?	Examples
Mechanical Wave	Yes	Sound, water, seismic waves
Electromagnetic Wave	No	Light, radio, X-rays, gamma rays

a. Mechanical Waves: These are waves that require a material medium (solid, liquid, gas) for transmission.

- **Examples:** Sound wave, wave on a rope, water waves.

b. Electromagnetic Waves: These are waves that do not require a material medium; can travel in a vacuum.

- **Examples:** Light, X-rays, radio waves.

(ii) Based on Particle Motion (Direction):

Type of Wave	Particle Direction	Wave Direction	Examples
Transverse	Perpendicular	Horizontal/Forward	Light, water waves
Longitudinal	Parallel	Horizontal/Forward	Sound, slinky spring

DETAILED TYPES OF WAVES

1. Transverse Waves: The particles of the medium move **perpendicularly** to the direction of wave propagation.

Features:

- Consist of **crests** and **troughs**
- Can travel through solids and the surface of liquids.

Examples:

- Light waves
- Water waves
- Waves on strings or ropes

2. Longitudinal Waves: The particles of the medium vibrate **parallel** to the direction of wave propagation.

Features:

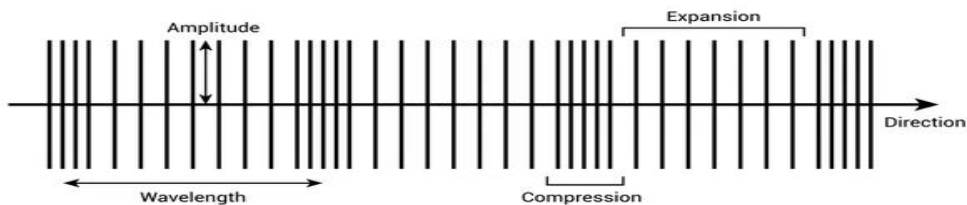
- Composed of **compressions** and **rarefactions**
- Travel through solids, liquids, and gases

Examples:

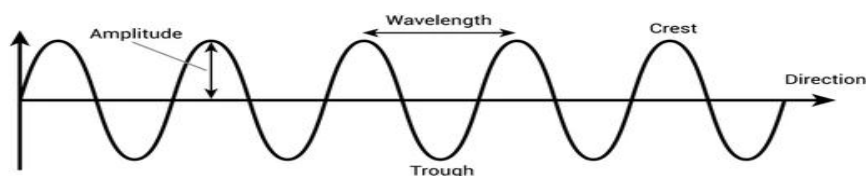
- Sound waves
- Slinky waves (along the spring)

DIFFERENCES BETWEEN TRANSVERSE AND LONGITUDINAL WAVES

Longitudinal Waves



Transverse Waves



Feature	Transverse Waves	Longitudinal Waves
Direction of vibration	Perpendicular to wave direction	Parallel to wave direction
Medium	Solids, surface of liquids	Solids, liquids, and gases
Features	Crest and Trough	Compression and Rarefaction
Example	Water waves	Sound waves

EVALUATION

1. Define a wave.
2. Differentiate between mechanical and electromagnetic waves with examples.
3. With the aid of diagrams, distinguish between transverse and longitudinal waves.
4. Give two examples each of:
 - Mechanical waves
 - Electromagnetic waves
 - Transverse waves
 - Longitudinal waves
5. A wave has a frequency of 5 Hz and wavelength of 2 m. Calculate its speed.

ASSIGNMENT

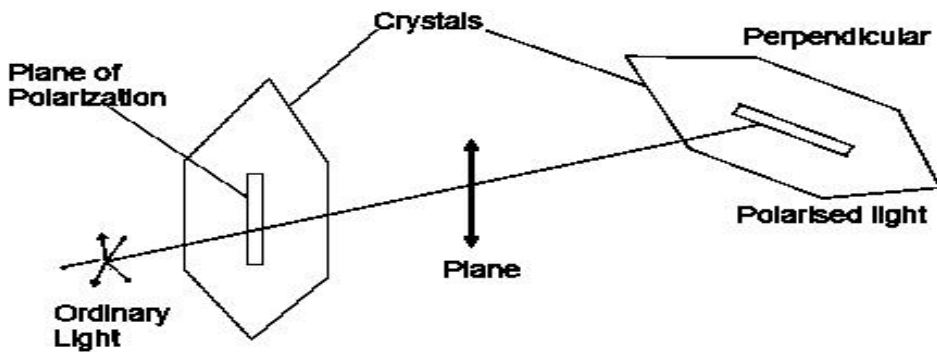
- **Draw and label:**
 1. One full transverse wave
 2. One full longitudinal wave (show compression and rarefaction)
- Research and list **5 applications** of electromagnetic waves in modern life.



WEEK 10 - PROPERTIES OF WAVES

PROPERTIES OF WAVES

- (i) **Reflection:** This is the bouncing back of a wave when it hits a surface.
Light reflecting off a mirror is a common example.
- (ii) **Refraction:** This is the bending of a wave as it passes from one medium to another.
This happens because the wave changes speed when it enters the new medium.
- (iii) **Diffraction:** Diffraction is the spreading of waves as they pass through an opening or around an obstacle.
- (iv) **Interference:** Interference occurs when two or more waves overlap, creating a new wave pattern. This can result in constructive interference (increased amplitude) or destructive interference (decreased amplitude).
- (v) **Stationary Wave:** These are wave obtained when two progressive waves of equal amplitude and frequency are travelling in opposite direction and combine together.
- (vi) **Polarization:** These involve the restriction of the vibration to only one plane.



Wave Equation:

$$Y = A \sin (wt - 2\pi x / \lambda)$$

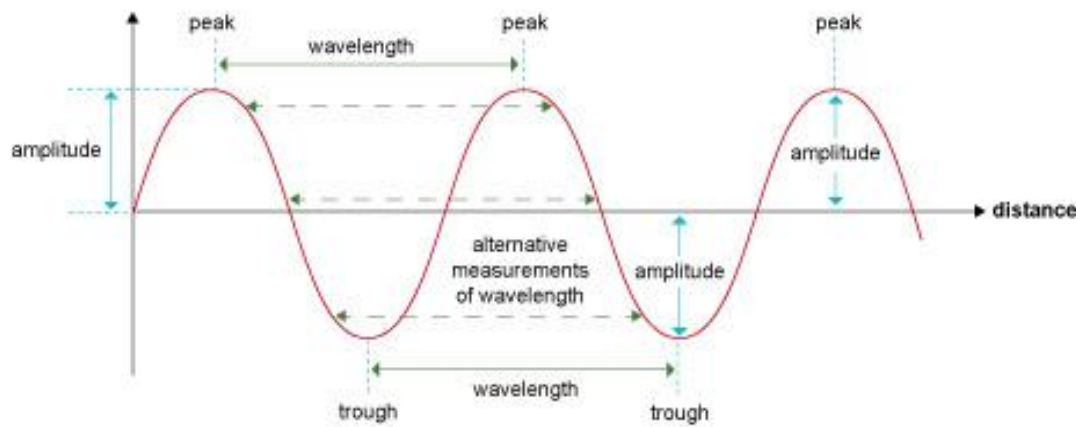
Where:

- A = Amplitude
- w = Angular frequency
- λ = Wavelength
- t = Time

- $x = \text{Position}$

GRAPHICAL REPRESENTATION OF WAVES

As you learn more about waves and the different situations in which they occur you will see that it is convenient to represent waves graphically. We can plot a wave as a graph in two ways: as a function of *distance* and as a function of *time*. It is important to appreciate the difference between them and the properties of the wave that are represented in both.



Mathematical Relationships between Frequency (f), Wavelength (λ) and Velocity (v):

This relationship is given by;

$$v = f\lambda$$

$$\text{Velocity} = \text{Frequency} \times \text{Wavelength}$$

Step 4: Work Examples

1. Surface waves travelling in deep water at 15ms^{-1} are incident at shallow water boundary. if the angles of incidence and refraction are 45° and 30° respectively, calculate the speed of the waves in shallow waters.

Solution

$$\text{index } (n) = \frac{\sin i}{\sin r} = \frac{\text{speed in deep water}}{\text{speed in shallow water}}$$

Substituting values,

$$\frac{\sin 45}{\sin 30} = \frac{15}{x}$$

$$x = \frac{15 \sin 30}{\sin 45} = 10.6 \text{ ms}^{-1}$$

2. Radio waves travel in air at $3.0 \times 10^8 \text{ ms}^{-1}$. If the waves enter water of refractive index $4/3$, calculate the speed of radio waves in water.

Solution

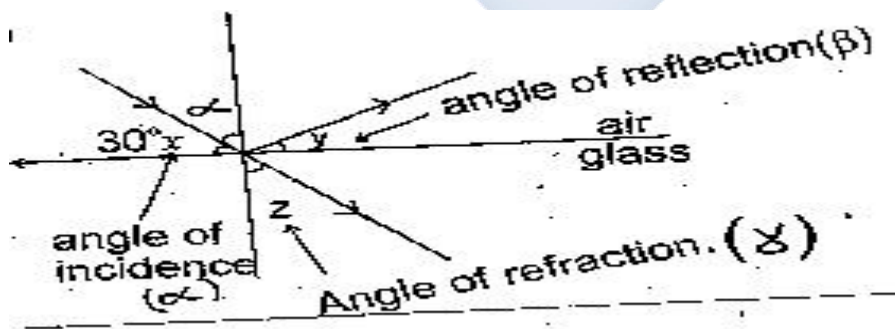
$$\text{Refractive index} = \frac{\text{speed of wave in air}}{\text{speed of wave in water}}$$

$$\text{Speed in water} = 3 \times 10^8 \times \frac{3}{4}$$

$$\frac{4}{3} = \frac{3 \times 10^8}{\text{Speed in water}}$$

$$= 2.25 \times 10^8 \text{ ms}^{-1}$$

3. A ray of light is incident on a surface of a rectangular glass prism of refractive index 1.5 as illustrated in the diagram below.



Solution

$$n = \frac{\sin i}{\sin r}$$

$$1.5 = \frac{\sin 60}{\sin r}$$

$$1.5 = \sin 60 / \sin r$$

$$r = \sin^{-1} 0.5774$$

$$r = 35^\circ$$

4. What is the frequency of a radio wave of wavelength 150 m if the velocity of radio waves in free space is $3 \times 10^8 \text{ ms}^{-1}$?

Solution

$$f = ? \quad v = 3 \times 10^8 \text{ ms}^{-1}; \lambda = 150 \text{ m}; v = f\lambda$$

$$f = \frac{v}{\lambda} = 3 \times \frac{10^8}{150}$$
$$= 2.0 \times 10^6 \text{ Hz}$$

EVALUATION QUESTIONS

- Define the following:
 - Amplitude
 - Wavelength
 - Frequency
 - Period
 - Wave speed
- Draw and label the parts of:
 - A transverse wave
 - A longitudinal wave
- A wave travels at a speed of 340 m/s with a frequency of 170 Hz. Calculate the wavelength.

4. Describe what happens to a wave during:
 - a) Reflection
 - b) Refraction
 - c) Diffraction
5. Differentiate between compression and rarefaction in longitudinal waves.

ASSIGNMENT

1. A wave has a period of 0.02 seconds. Calculate its frequency.
2. Describe three real-life examples of wave interference.
3. Discuss how wave diffraction affects sound in a building.

